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ABAP Enhancements

Deloitte Technology Academy (DTA)

Agenda

Topic	Descriptions	Duration
Enhancements	Changing SAP Standard Code, and Overview of Enhancements	8 hours
Data Dictionary Enhancements	Enhancements to the Data Dictionary	
Business Transaction Events (BTE)	What Are Business Transaction Events (BTE), Different Kinds of Interfaces, and Using and Finding BTEs	
Modifications	Overview of Modifications, and User Exits	
Customer Exits	Function Module Exits, Screen Exits, and Menu Exits	
Business Add-Ins (BADI)	Components, Process Flow Diagram, Calling Programs, BADI Function Code Overview and Restrictions, BADI Function Code Processing in the Program, Defining BADI, and Implementing BADI	

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Learning Objectives

By the end of this session, you will be able to:

- Describe how to change SAP standard code
- Describe enhancements
- List the types of enhancement
- Enhance tables using append and include
- Explain business transaction events and software delivery process
- Describe the process flow diagram of BTE functions and how to implement it
- Implement customer exits including function module exits, screen exits, and menu exits
- Describe how to implement business add-ins (BADI)
- Explain the process flow diagram of BADI
- Explain BADI function codes and define BADI

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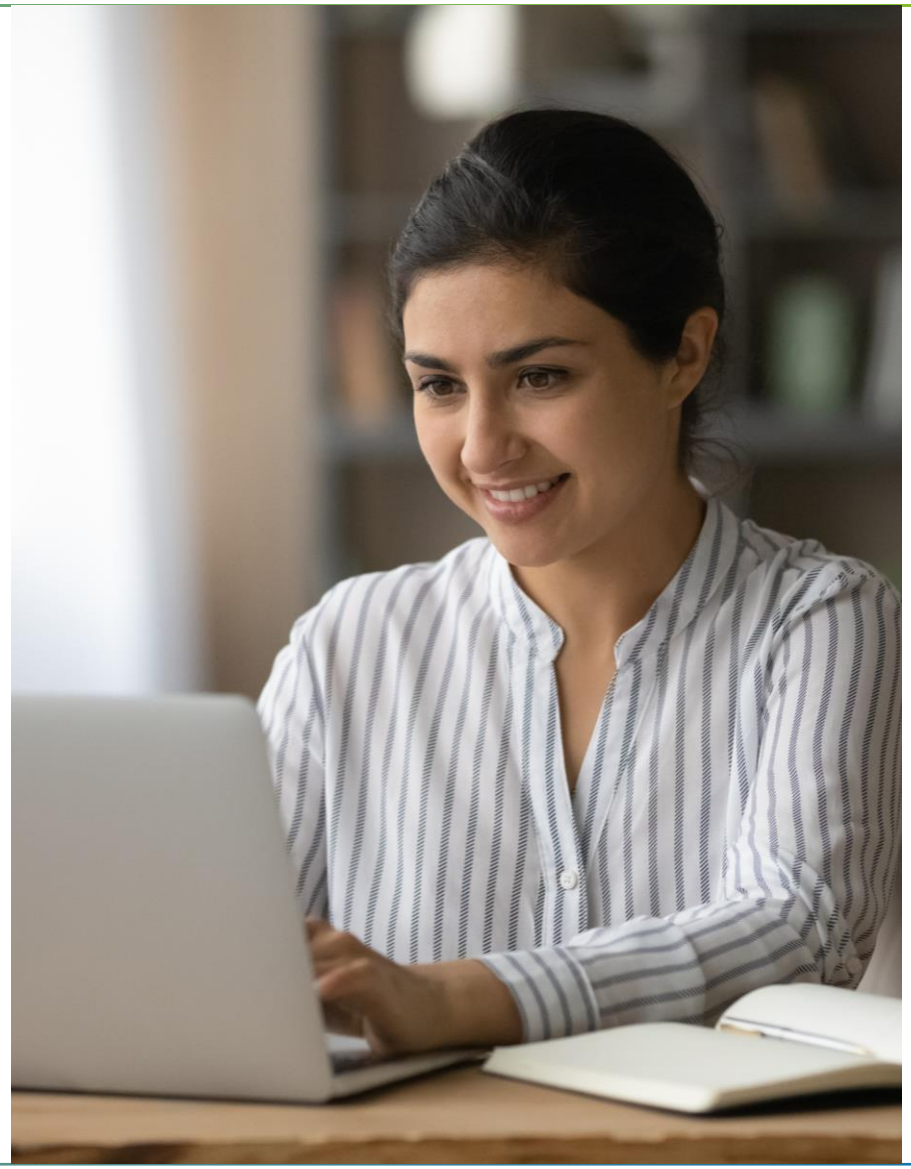
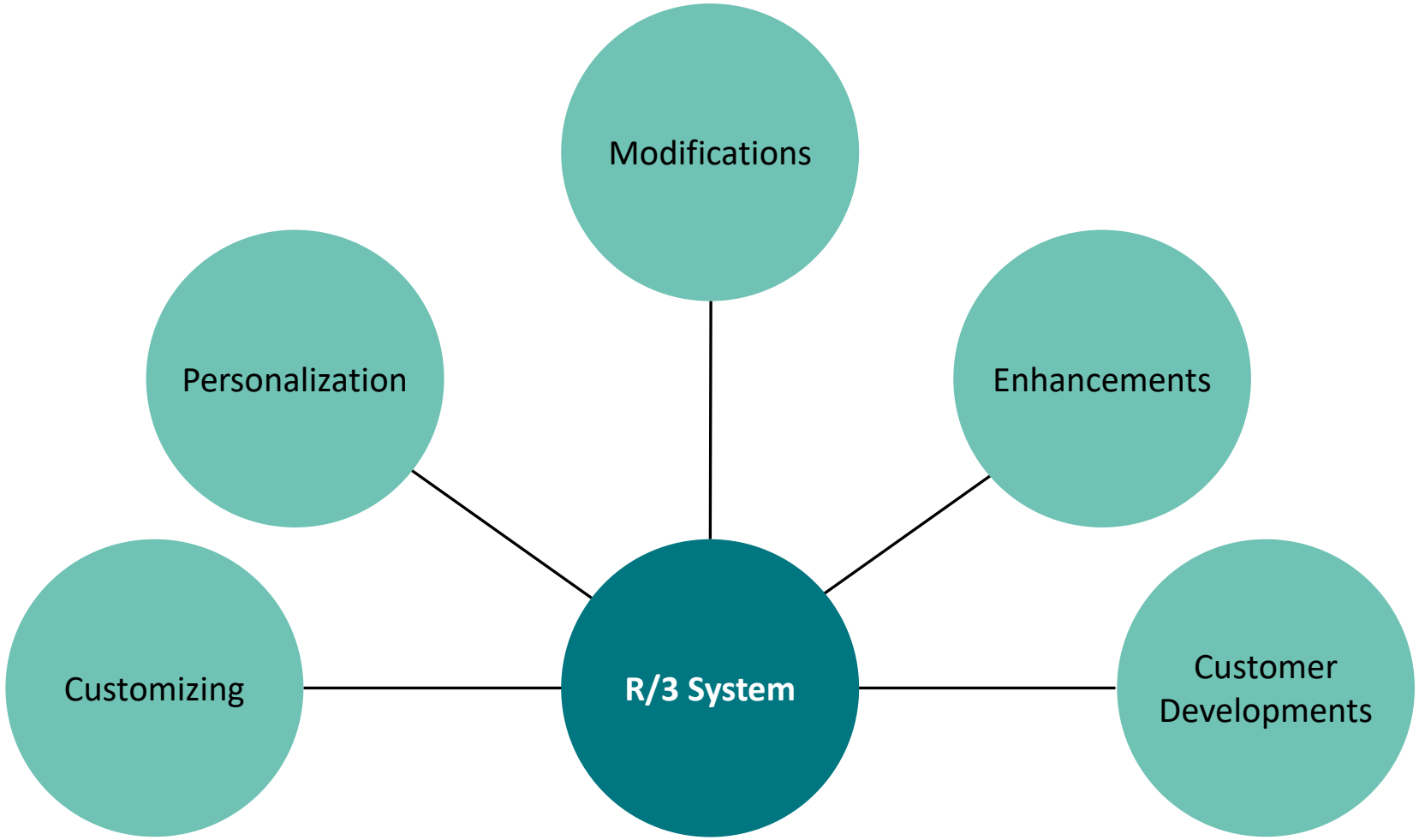
Enhancements



Changing SAP Standard Code

You can adjust the R/3 System to meet your needs in the following ways:



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Adjusting the R/3 System to Meet Your Needs

Customizing	This means setting up specific business processes and functions for your system according to an implementation guide. The need for these changes has already been foreseen by SAP and an implementation procedure has been developed.
Personalization	This means making changes to certain fields' global display attributes (setting default values or fading fields out altogether), as well as creating user-specific menu sequences.
Modifications	These are changes to SAP repository objects made at the customer site. If SAP delivers a changed version of the object, the customer's system must be adjusted to reflect these changes. Prior to release 4.0B, these adjustments had to be made manually using upgrade utilities from release 4.5A, this procedure has been automated with the modification assistant.
Enhancements	This means creating repository objects for individual customers that refer to objects that already exist in the SAP repository.
Customer Developments	This means creating repository objects unique to individual customers in a specific namespace reserved for new customer objects.

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Overview Diagram—Changing the Standard

R/3 Business Applications
(The SAP Standard)

Personalization

- Activity groups
- Transaction variants

Modification

- Non-registered modification
- Assisted modification
- User exits

Enhancement

- ABAP Dictionary
  - Tables
  - Field exits
- Customer exits
  - Function module exits
  - Menu exits
  - Screen exits
- Business transaction events
- Business add-ins

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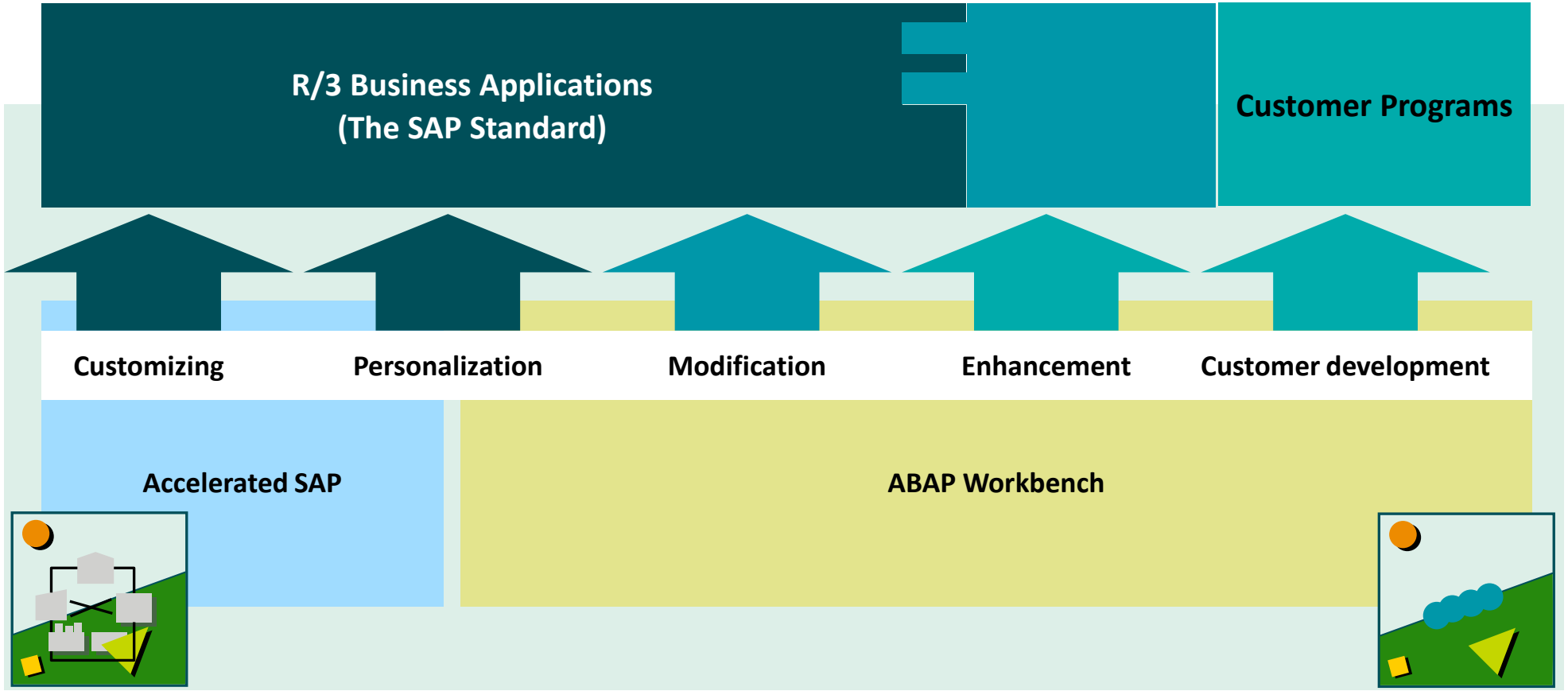
Change Levels



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Why Enhancements?

Implications of changing SAP Standard Code:

- Dependency of objects across modules
- During upgrades changes to standard SAP code are overwritten
- No support from SAP in event of issues encountered from modified SAP code

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Enhancing Your System

You can enhance your system at the following levels:

In ABAP dictionary tables or structures (table enhancement)

GUI interfaces (menu exit)

ABAP programs (function module exit)

Screens by inserting a sub-screen in an area specified by SAP (screen exit)

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Comparison of Enhancement Techniques

Enhancement Options:

	Customer Exits	Business Transaction Events	Business Add-Ins
Source Code	+	+	+
Menus	+	-	+
Reusable	-	+	+
Filter Specific	-	+	+

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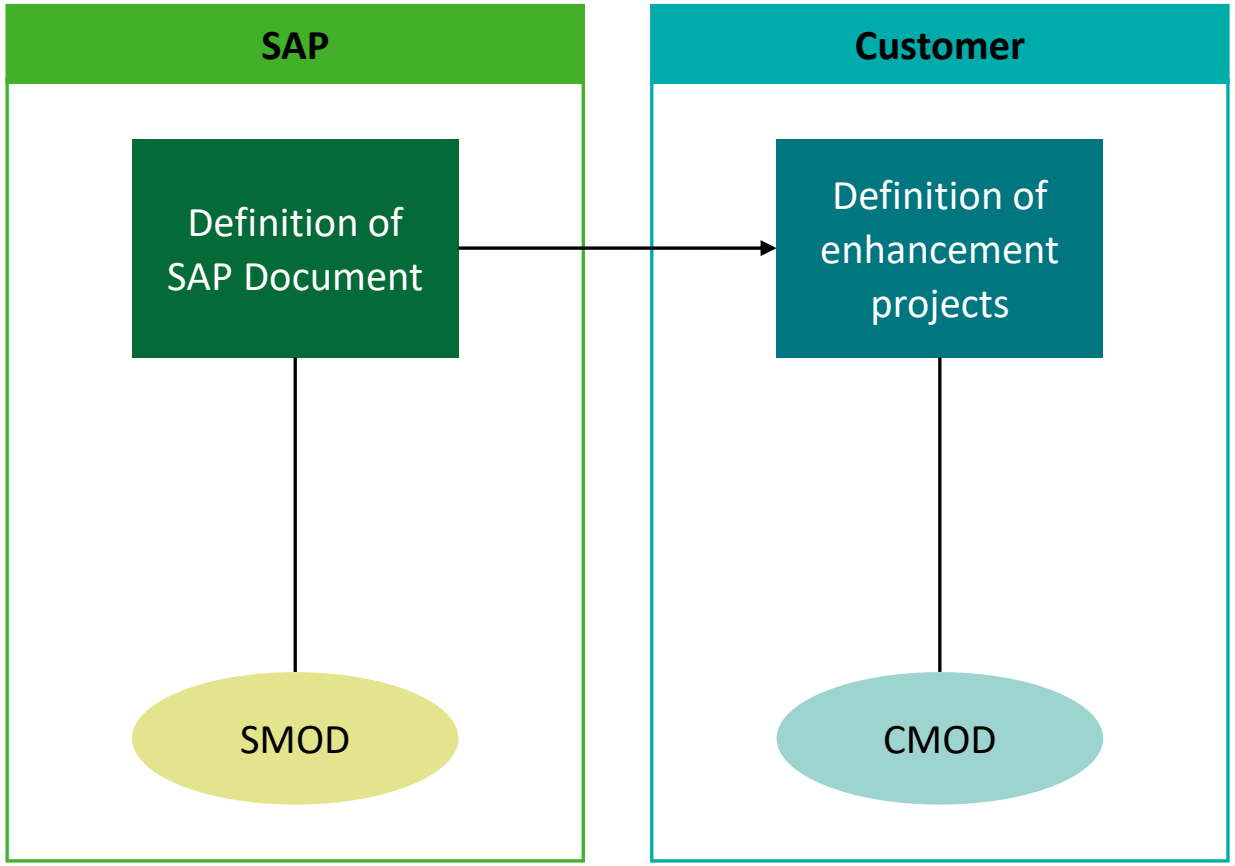
Customer Enhancements Projects

- The SAP application programmer creates SAP enhancements from the function module exits, menu exits, and screen exits. A management function is provided for this purpose (transaction code SMOD).
- Customers are given a catalog containing an overview of existing SAP enhancements. They can then combine the SAP enhancements they want into an enhancement project using transaction CMOD.



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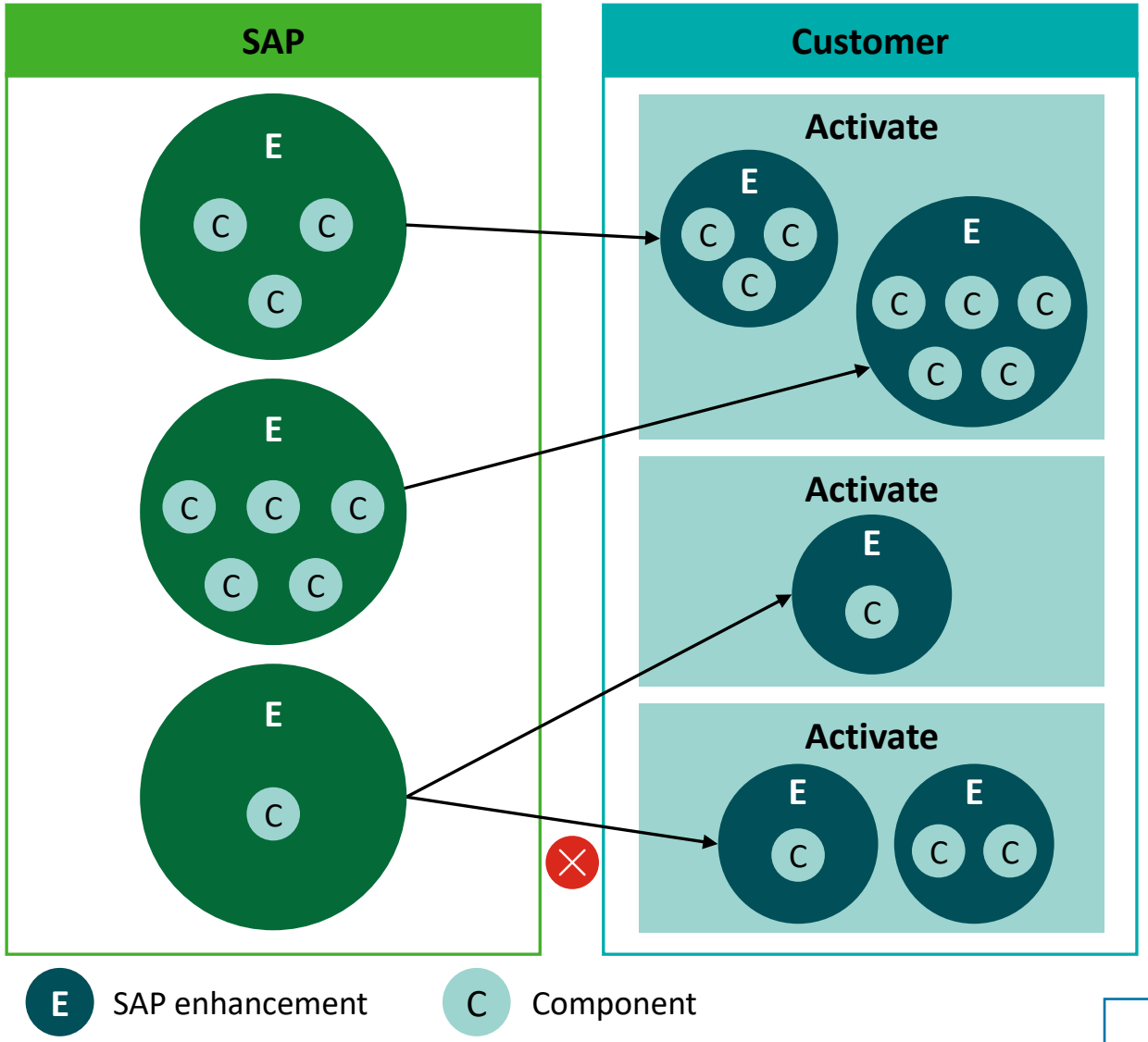
Enhancements and Enhancement Projects

- SAP enhancements are made up of component parts. These components include function module exits, menu exits, and screen exits. A specific component may be used only once in a single SAP enhancement (this guarantees the uniqueness of SAP enhancements).
- Customer enhancements projects consist of SAP enhancements. Each individual SAP enhancement may be used only once in a single customer enhancement program (this guarantees the uniqueness of a customer project).



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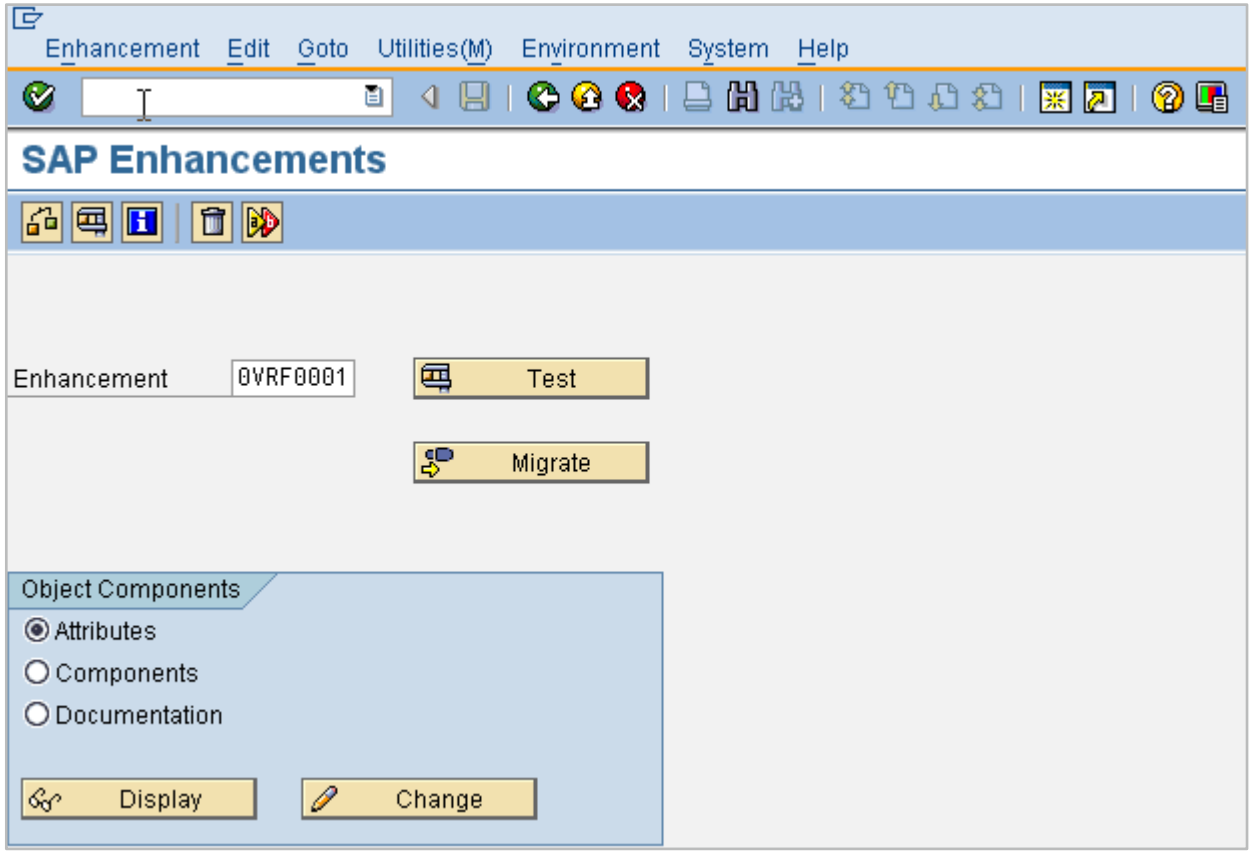
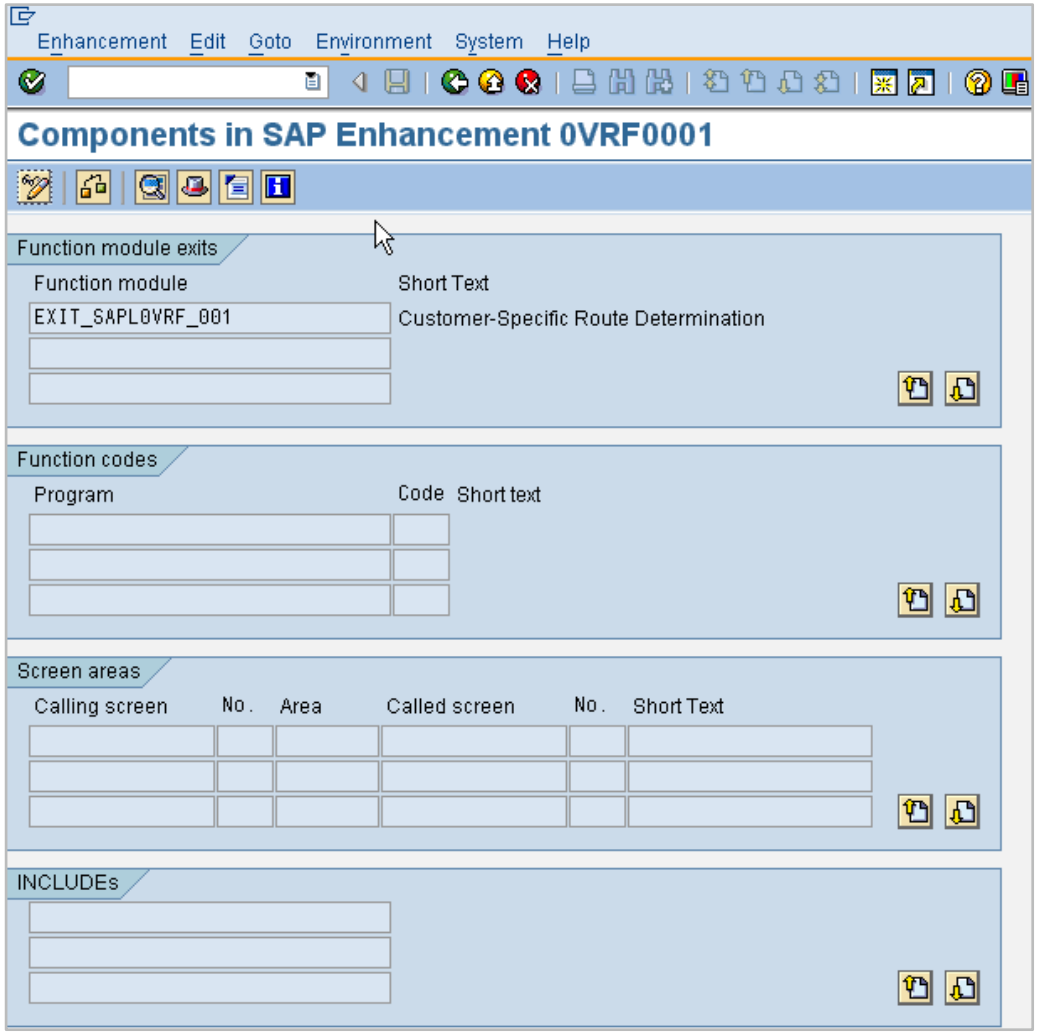
SAP Enhancements (SMOD)





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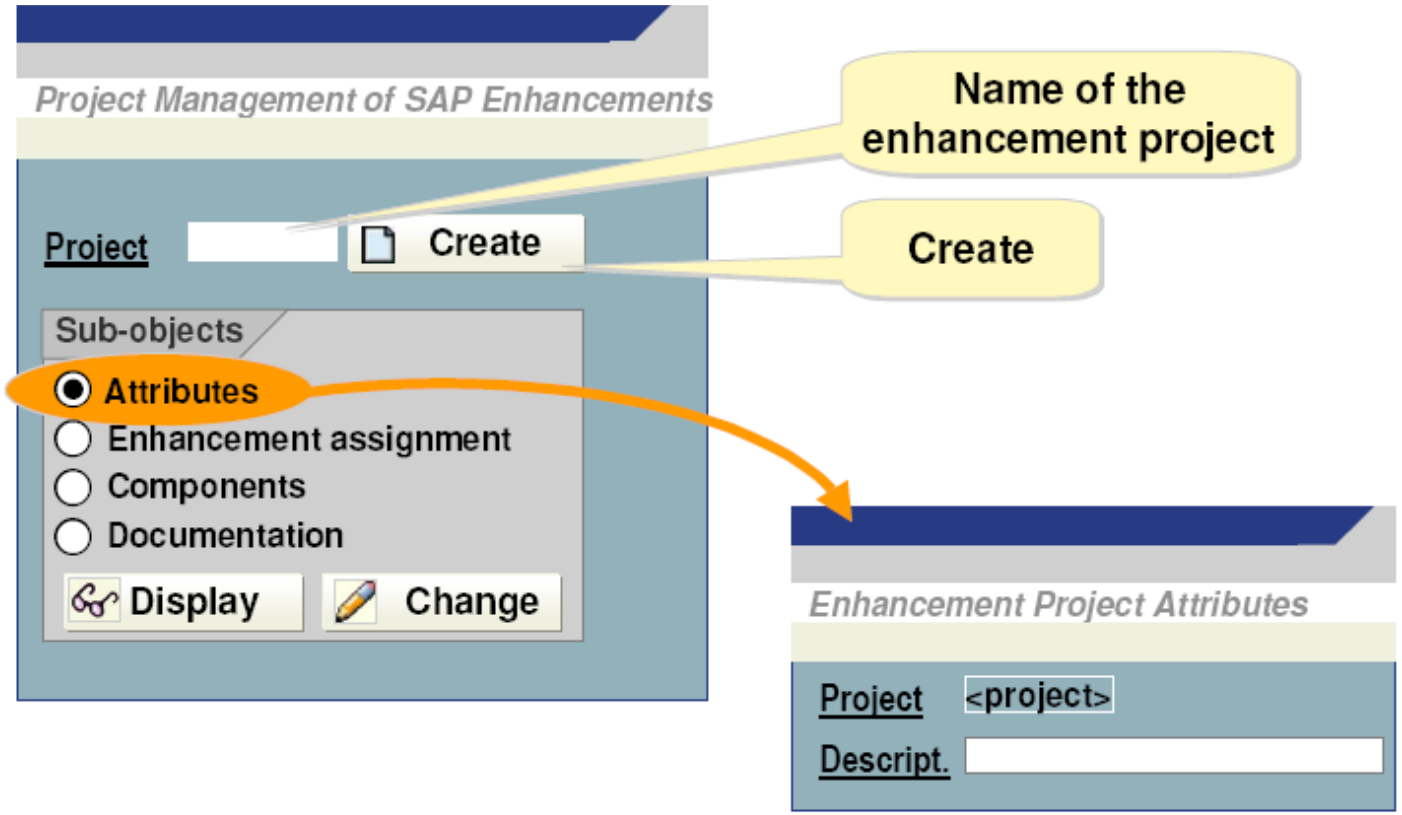
Creating Customer Enhancement Projects



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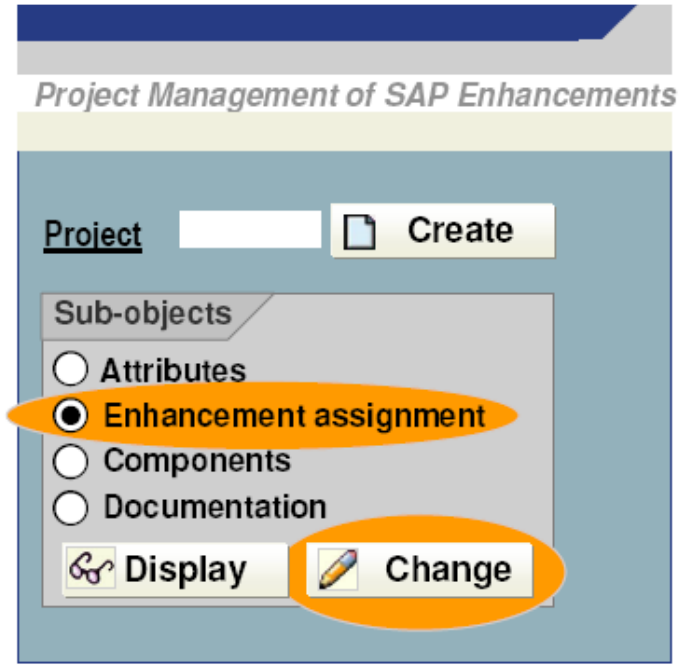
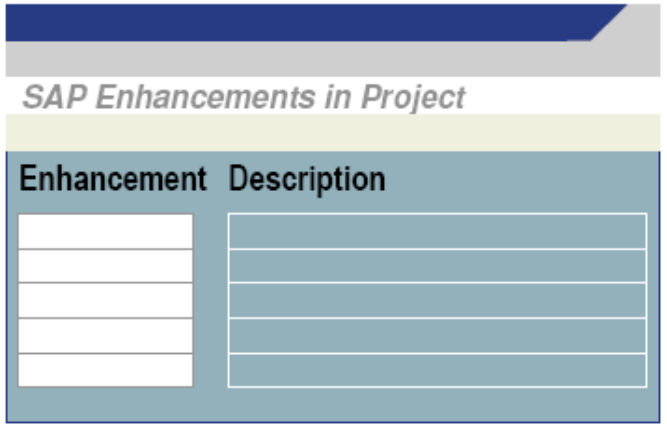
Assigning SAP Enhancements Customer Projects





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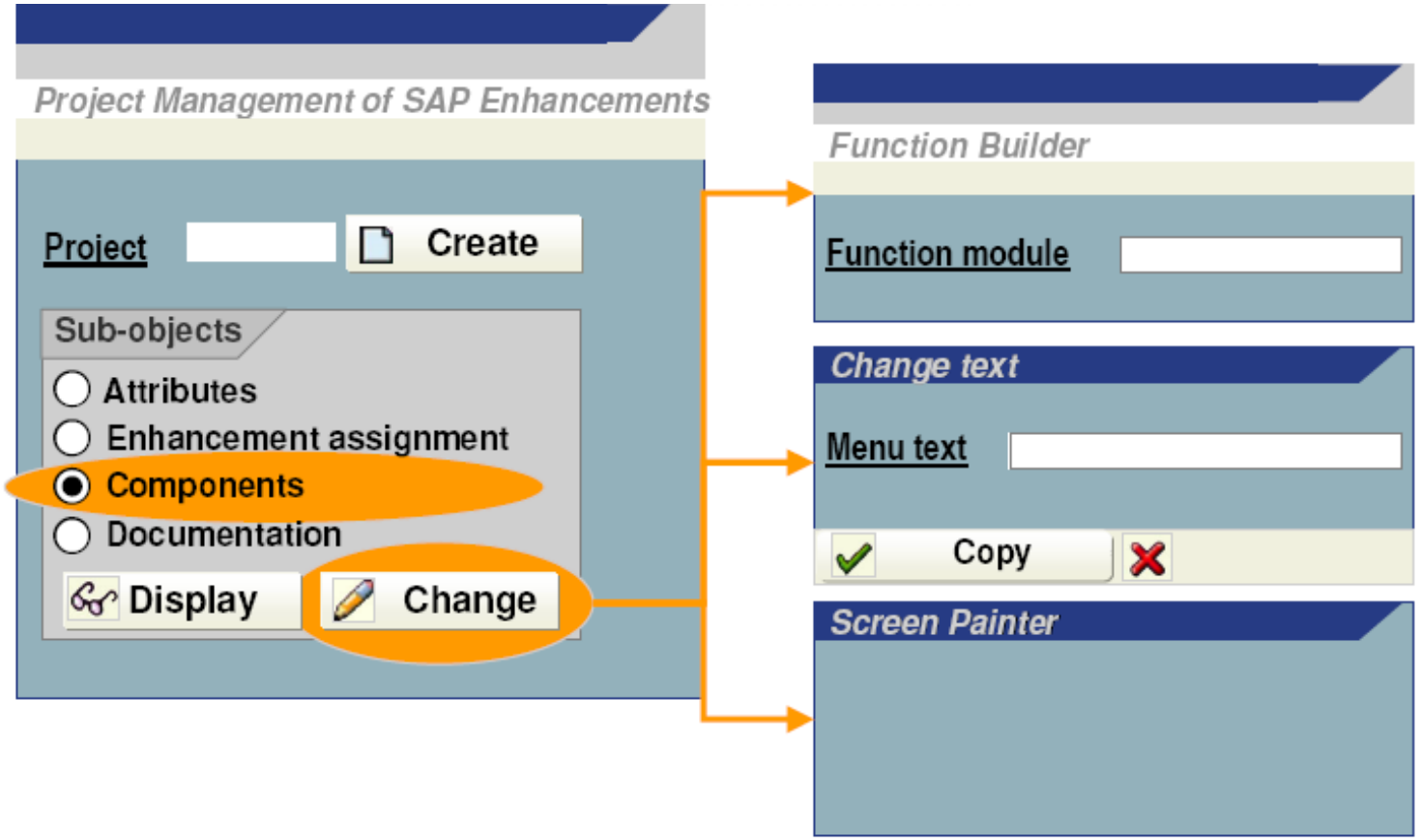
Editing Components



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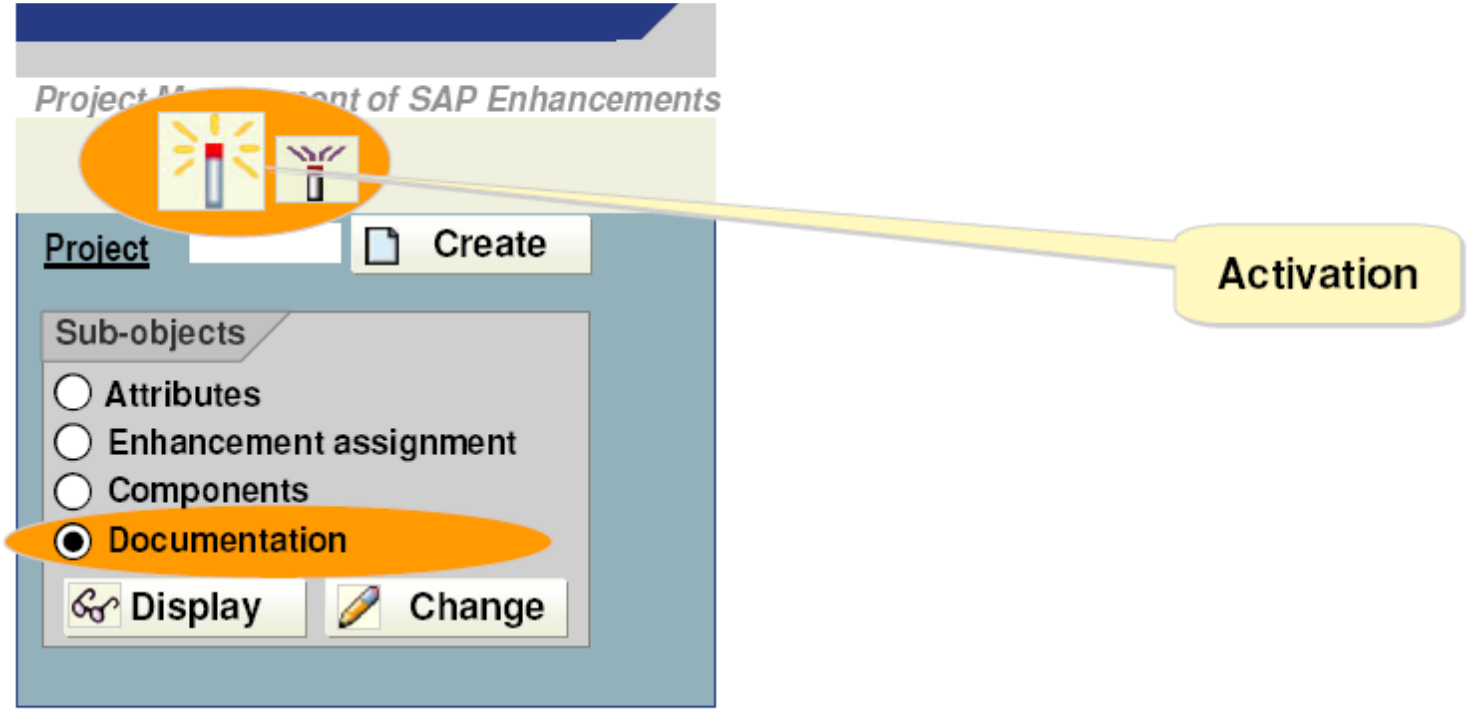
Activating Enhancement Projects



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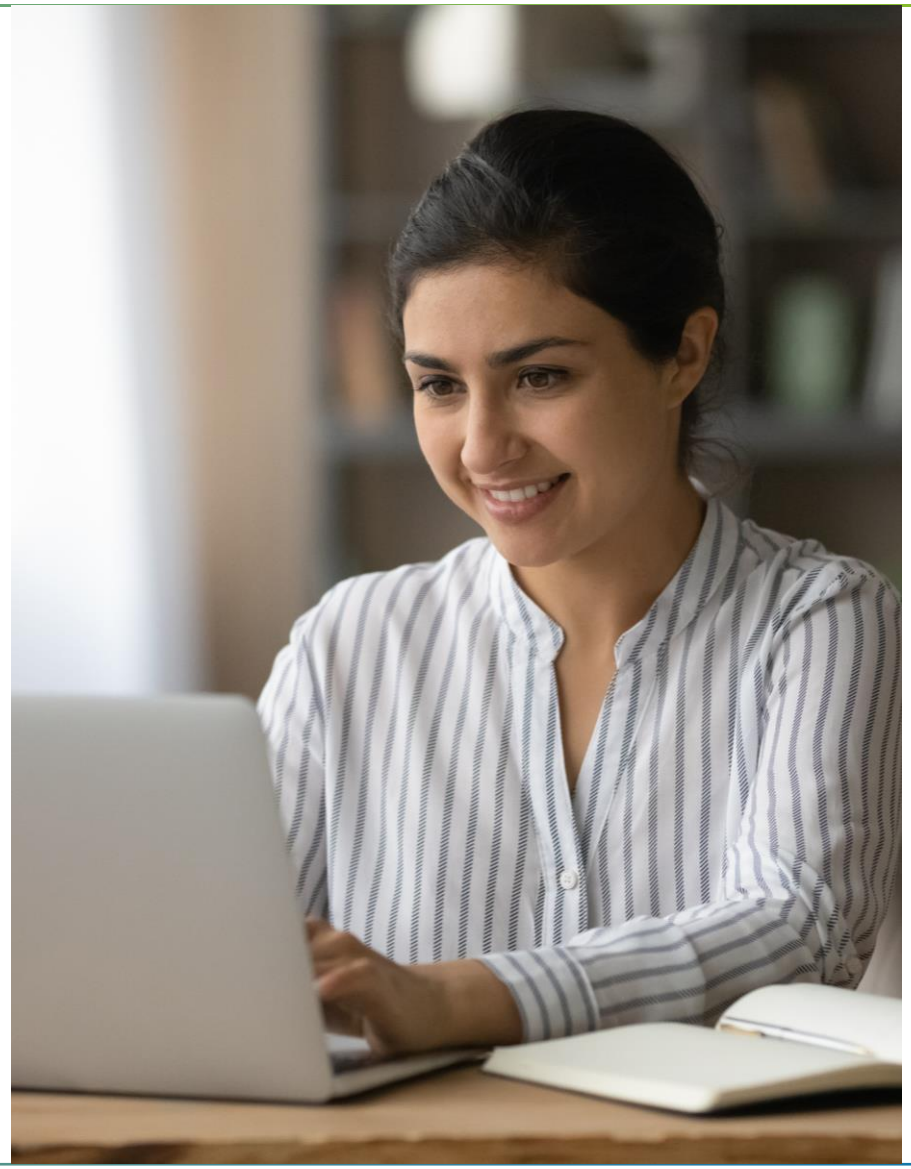
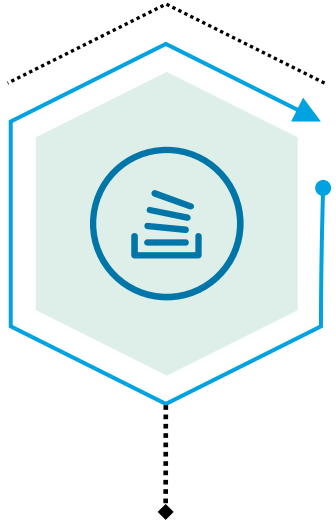
Data Dictionary Enhancements



SAP Provides Two Ways to Enhance Tables and Structures with Fields



Customizing includes ("CI includes")

- In contrast, `CI_includes` allows you to use the same structure in multiple tables. The include statement must already exist in the SAP table or structure.

- Append structures allow you to attach fields to a table without having to modify the table itself. An append structure can only belong to exactly one table.

Append structures

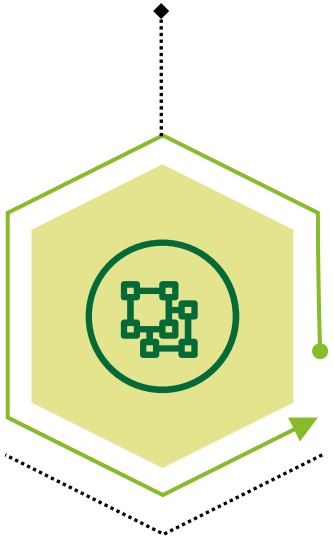


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Table Enhancements

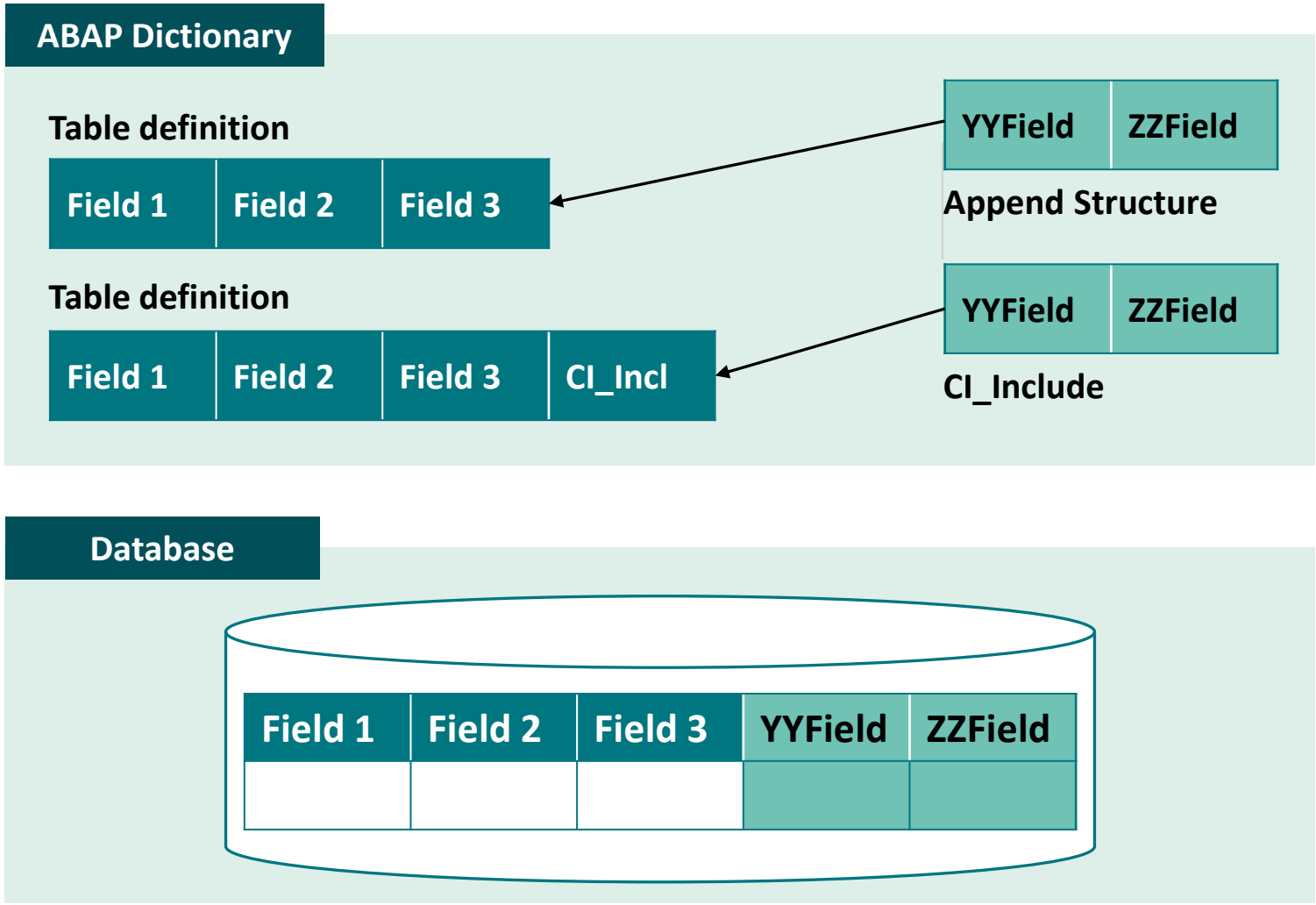

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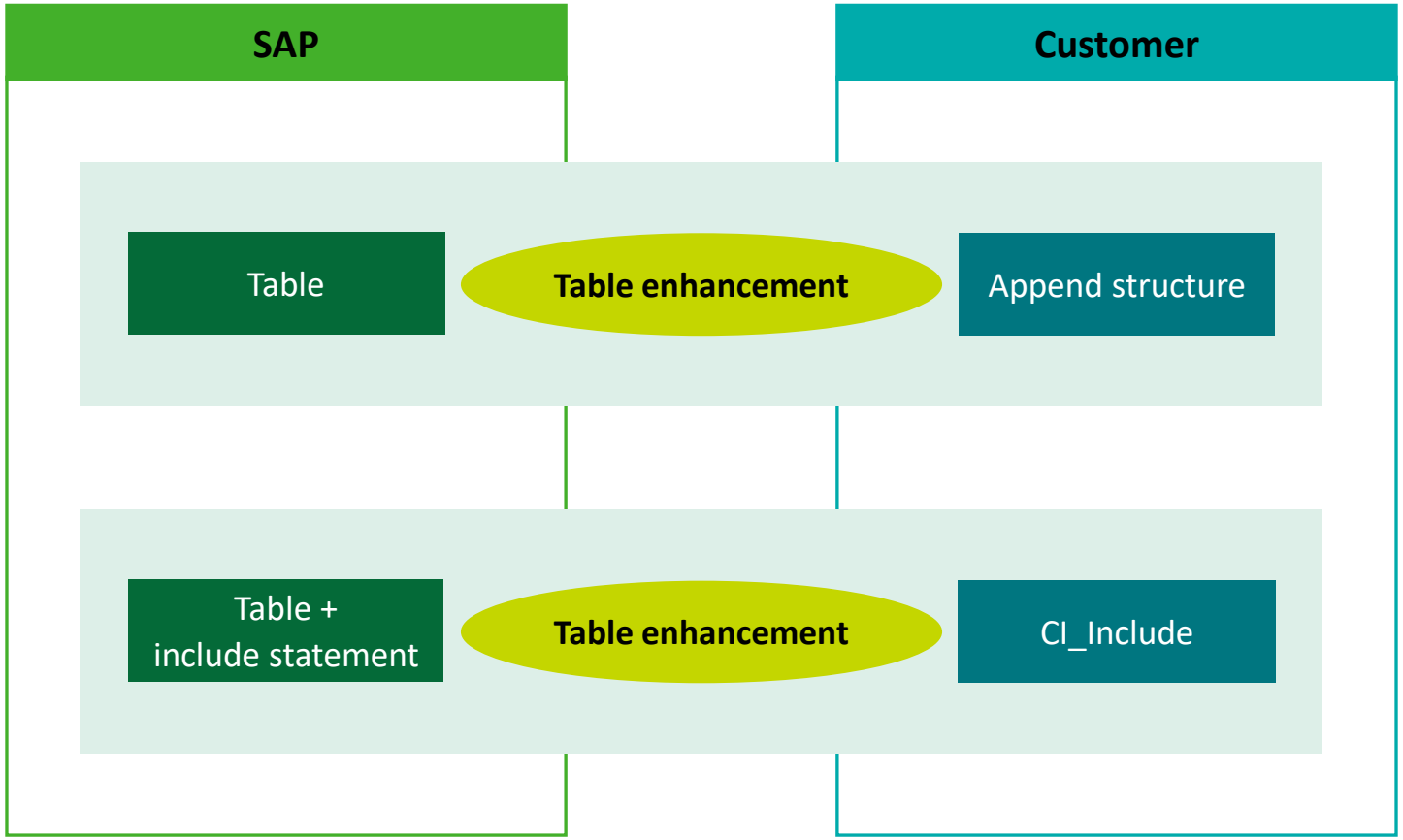
Table Enhancements—SAP and the Customer



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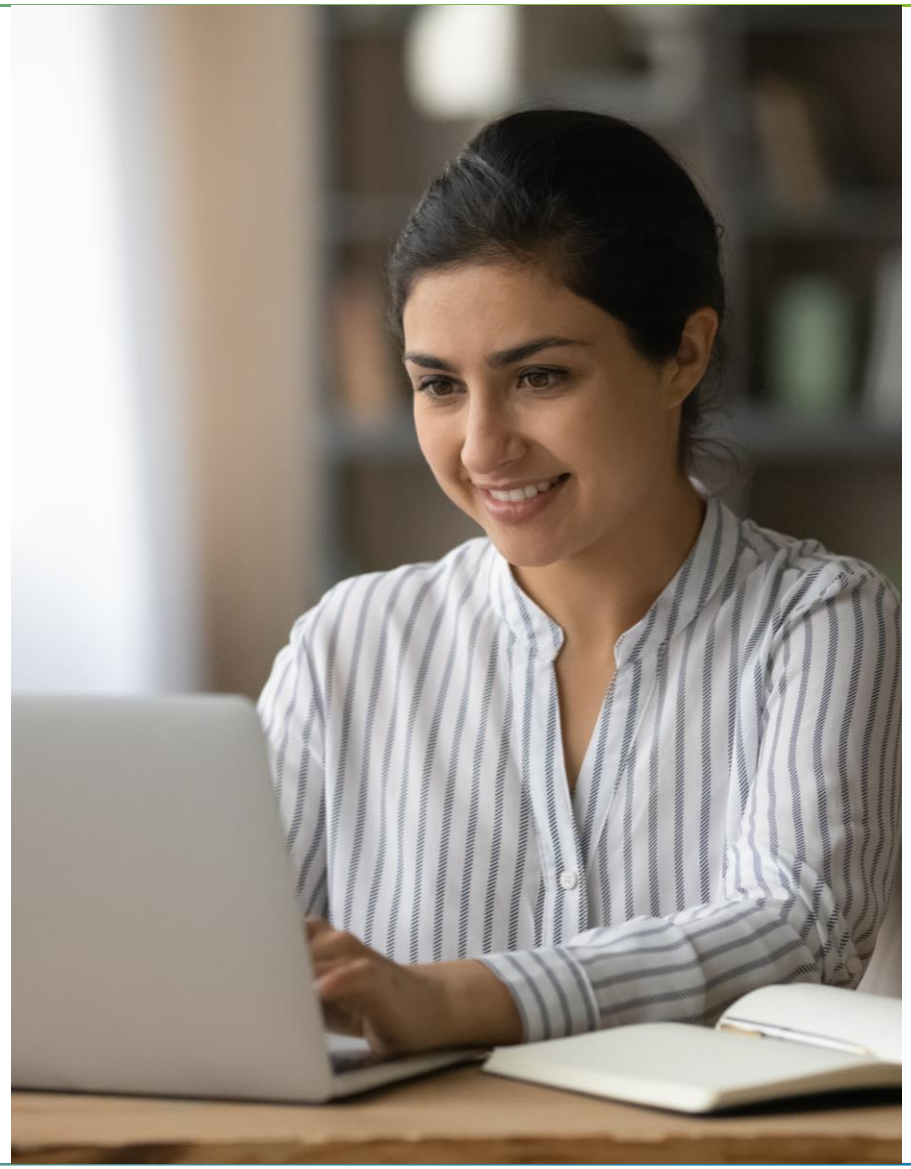
Business Transaction Events (BTE)



Business Transaction Events (BTEs)

- Are enhancements that were developed for the Financial Accounting (FI) component of R/3
- Have a predefined interface allowing customers access
- Have one of the following types of interfaces:
  - Publish and subscribe
  - Process interfaces

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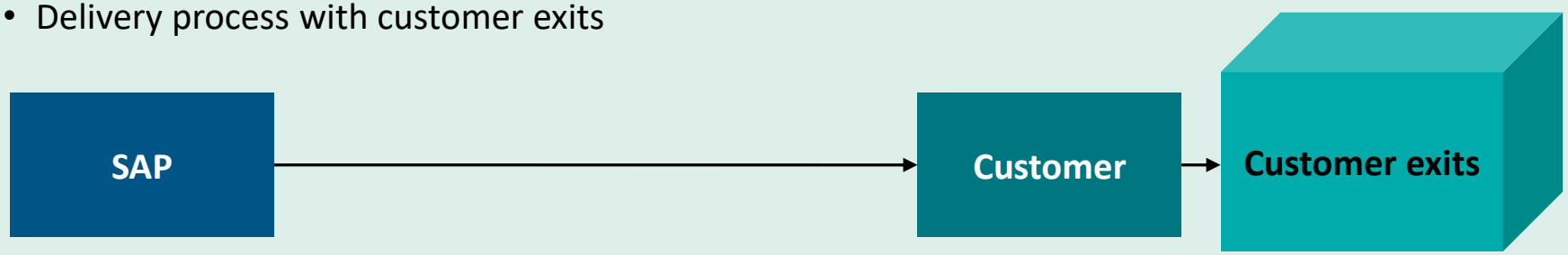
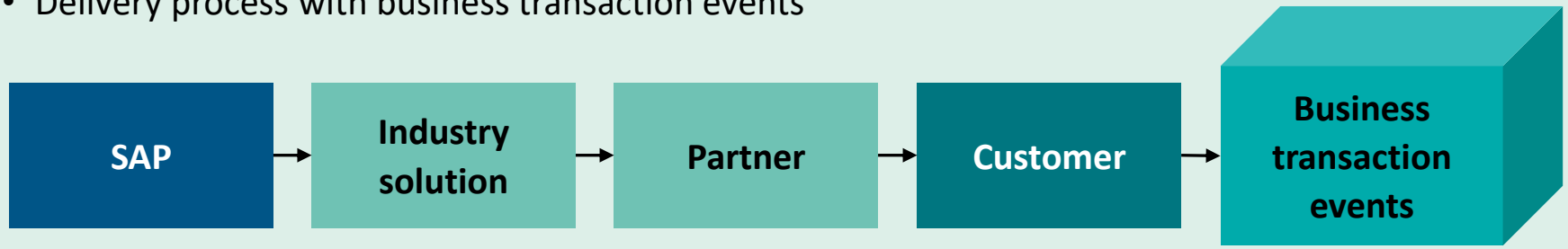
BTE—Software Delivery Process

- Delivery process with customer exits



- Delivery process with business transaction events



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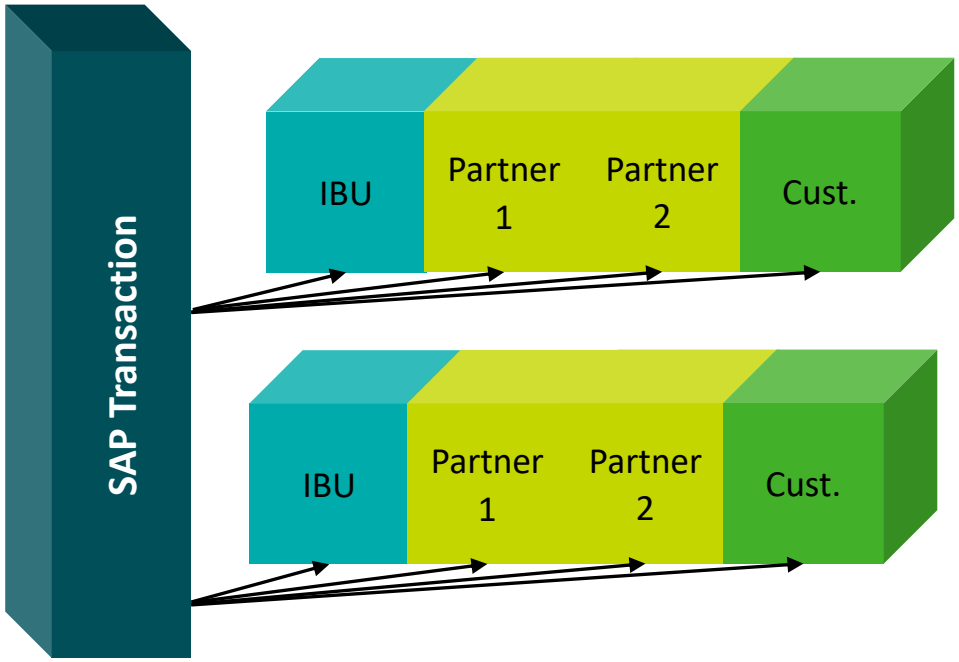
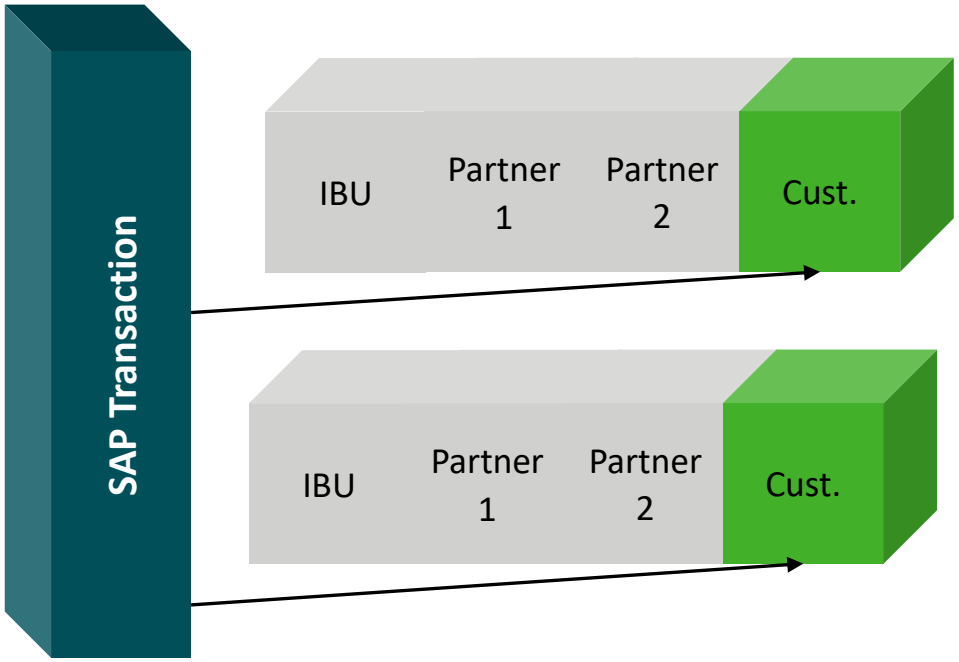
Publish and Subscribe Interfaces; Process Interfaces

- P/S interfaces



Triggers one or more additional operations

- Process interfaces



Process exactly one add-on component

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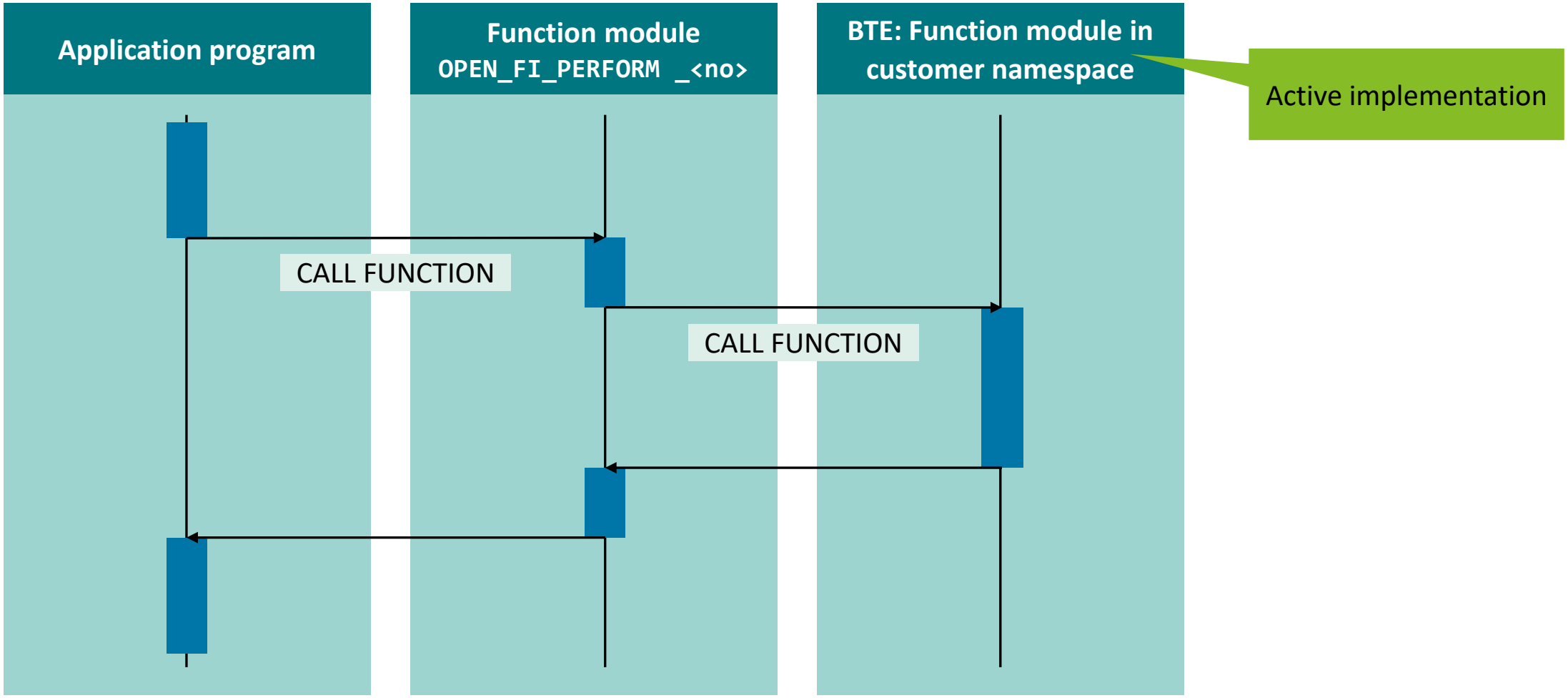
Business Transaction Event Functions—Process Flow Diagram



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Business Transaction Events—Calling Programs

```
REPORT <program_using_bte>.
START-OF-SELECTION.
…
```

```
CALL FUNCTION
 'OPEN_FI_PERFORM_00001350_E'
    EXPORTING
    <I_variables>
```

1

```
FUNCTION-POOL <sap_fp>.
…
FUNCTION open_fi_perform_00001350_e.
…
LOOP AT itab into wa.
```

2

```
CALL FUNCTION wa-fb_name
   EXPORTING
    <i_variables>
```

```
ENDLOOP.
…
ENDFUNCTION.
```

	FM_NAME
00001350	Z_FUBA001
00001350	/ABC/FUBA002
00001350	/CUSNR/FUBA003

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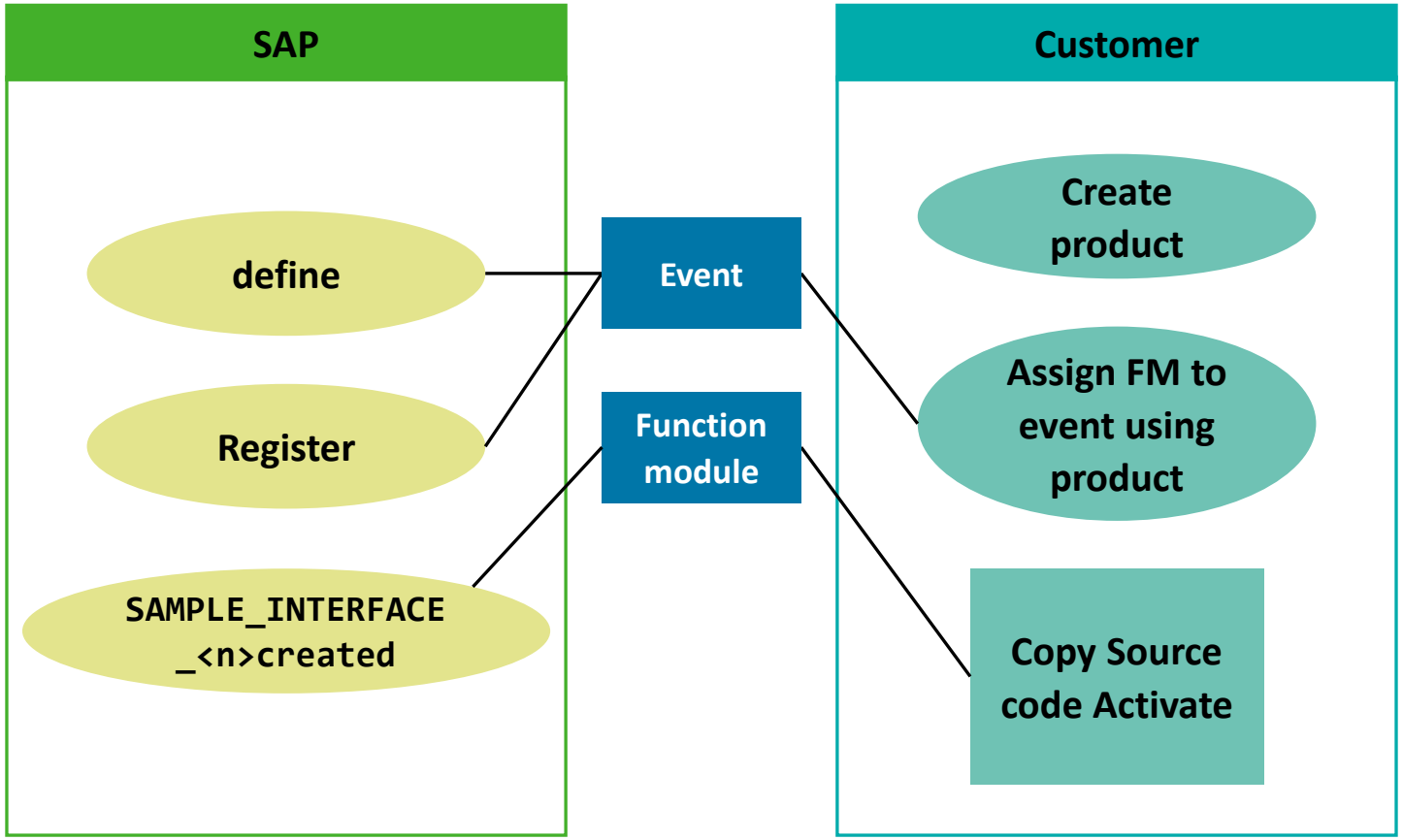
Business Transaction Events—SAP and the Customer



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Finding a Business Transaction Event

- **Directly in application:**

Business Transaction Events

System → Status
→ Double-click on program name
Find character string…

OPEN_FL_PERFORM

- `Global in program <prog>`
  - SAP Reference IMG

Financial Accounting
  - Basic settings
  - Use BTE

Environment
  - Info system

- **Using tools**

IMG

For example:

Financial Accounting
  - Basic Settings
    Use BTE

Environment
- Info system
  - Choose application component
  - Documentation

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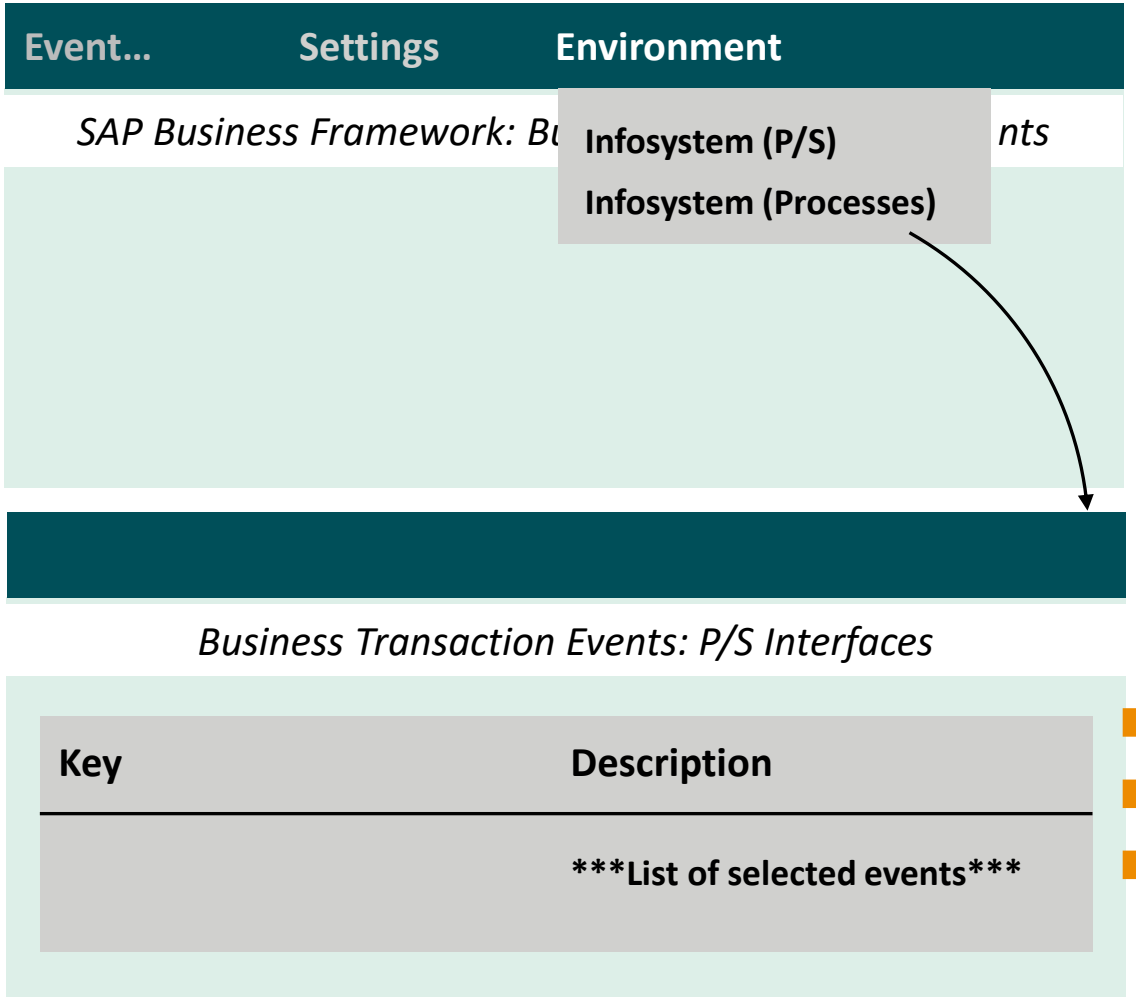
BTE—Documentation



- Model FM
- Interface
- Documentation

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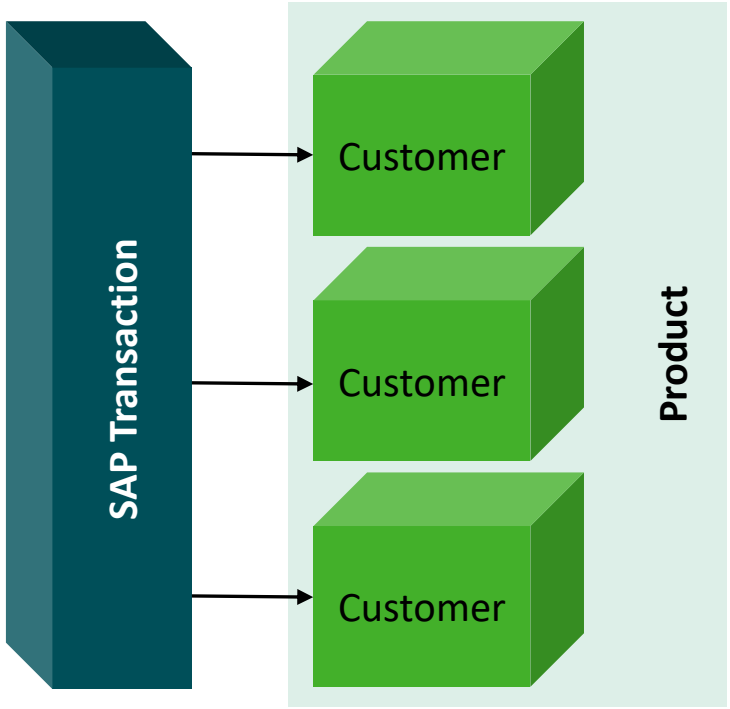
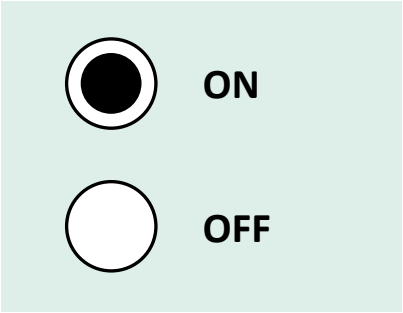
BTE—The Product Function



- The product groups enhancements together
- Can be activated or deactivated as a whole



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Implementing a BTE

Function Builder

Function module SAMPLE_INTERFACE_<N>_E → Z_*

- Copy the function module to the customer namespace
  - Source code
  - Activation

Event…	Settings	Environment
SAP Business	…	*s transaction events*
	Products ▶	**of a partner**
	P/S modules	**of a customer**
	Process modules	

- Create product

Event…	Settings	Environment
SAP Business	…	*s transaction events*
	Products	
	P/S modules ▶	…of a SAP user
	Process modules	…of a partner
		…of a customer

- Assign function module

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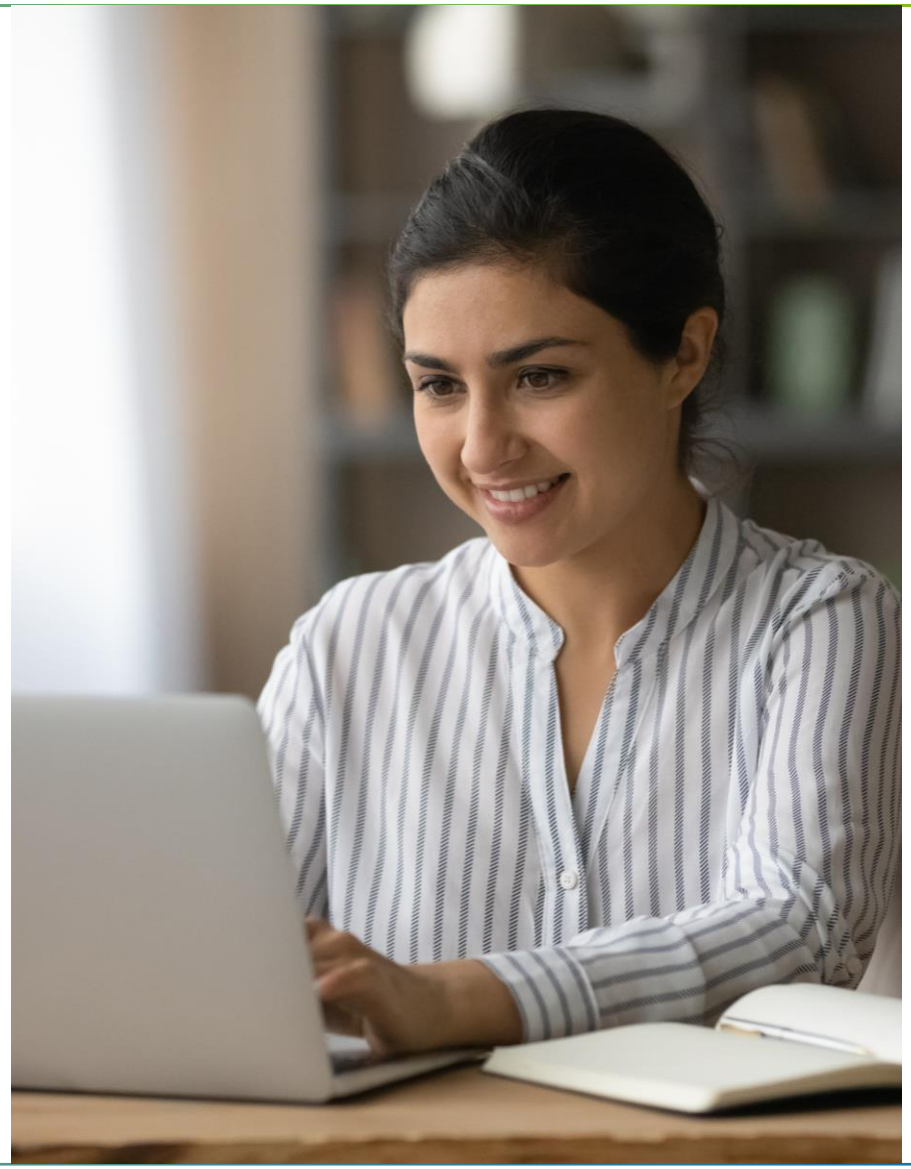
Modifications



What Are Modifications?

- Modifications are changes to SAP objects in customer systems. They are executed with the help of user exits (these are subroutines reserved for customers that have been inserted in objects in the SAP namespace).
  Example: MV45AFZZ (for sales processes) 'hard-coded' at various points within SAP repository objects. Customer developments are programs developed by customers that can call SAP repository objects.

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Modifications

Any change that you make to your system to an object that has been delivered by SAP is known as a modification.

Modifications can lead to complications at upgrade. When SAP delivers a new version of the object, you must decide whether the new object should be used, or whether you want to continue using your old object.

SAP

SAP program

```
PROGRAM … .
…

*Original Coding..
…
…
…

…
*Original Coding..
..
```

Customer

SAP program

```
PROGRAM … .
…

*Original Coding..
…
…
…
*Code changed or added by
the customer
…
*Original Coding..
..
```


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User Exits Overview

User exits are a type of system enhancement that was originally developed for the R/3 SD (Sales and distribution) module. They are:

- Empty subroutines that SAP Developers have provided for you to fill them with your own source code
- Also known as FORM EXITs
- Generally collected in includes and attached to the standard program by SAP

All User exits start with the word USEREXIT_.

Syntax:

```
FORM USEREXIT_XXXX....
 INCLUDE ZZUSEREXIT.......
  ENDFORM.
```


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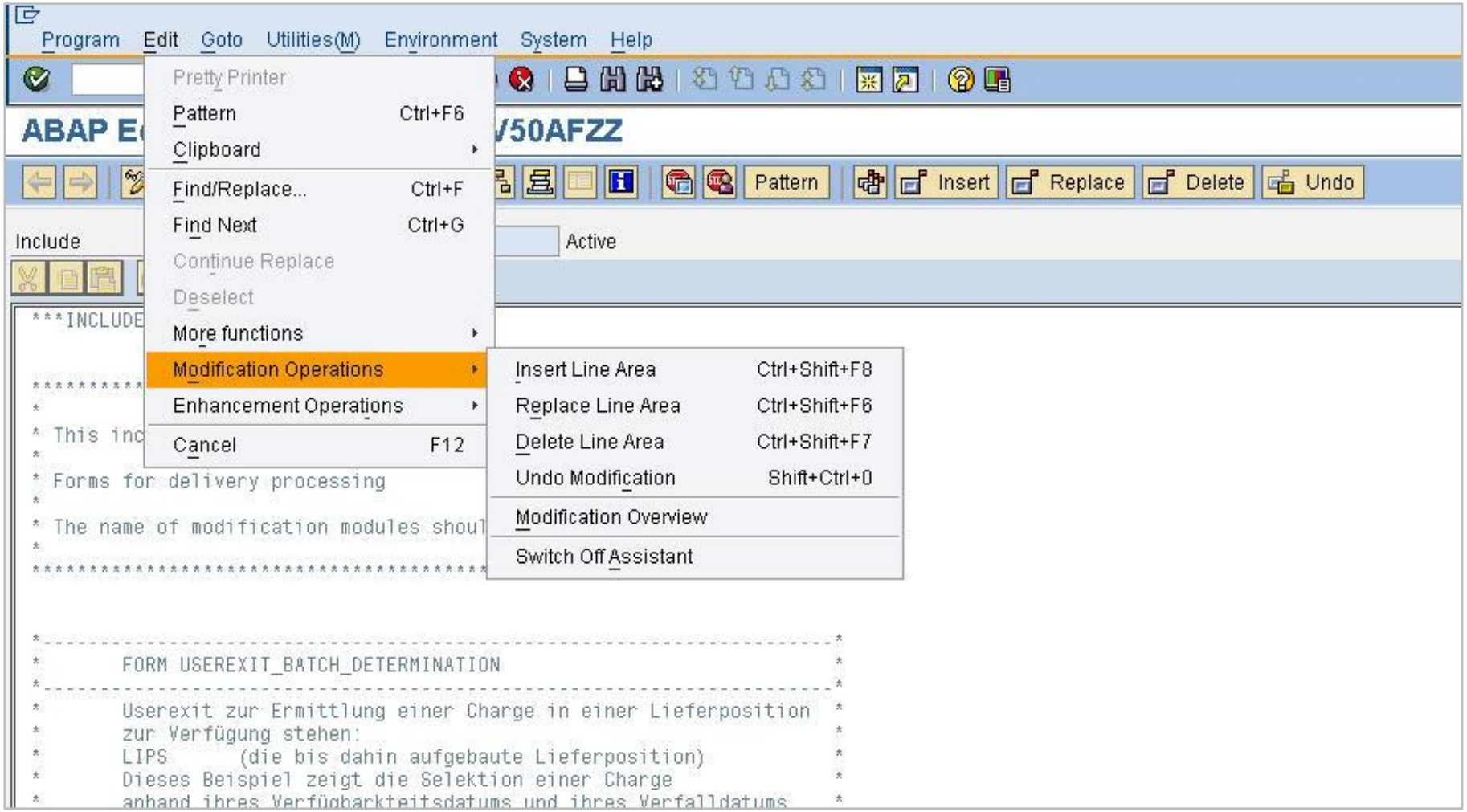
User Exits



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User Exit

```
FORM USEREXIT_LGORT_DETERMINATION.
*       This userexit can be used to fill the field LIPS-LGORT,       *
*       when a delivery is created                                    *
*       This form is called in the form LIPS-LGORT_ERMITTELN in the   *
*       Include FV50XFLP                                              *
*       you can use LIKP, some fields of LIPS, MAAPV, ...             *

* LIPS-LGORT = ........
*{   INSERT         &$&$&$&$                                          1
*
*}   INSERT

ENDFORM.
```

```
FORM USEREXIT_LGORT_DETERMINATION.
*       This userexit can be used to fill the field LIPS-LGORT,       *
*       when a delivery is created                                    *
*       This form is called in the form LIPS-LGORT_ERMITTELN in the   *
*       Include FV50XFLP                                              *
*       you can use LIKP, some fields of LIPS, MAAPV, ...             *

* LIPS-LGORT = ........
*{   INSERT         &$&$&$&$                                          1
* Data declaration
data: lv_vbeln type vbak-vbeln.
*}   INSERT

ENDFORM.
```


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User Exit—Example

```
***INCLUDE MV45AFZB .

* * * * * * * * * * * * * * * * * * * * * * * * * * * * * * *
* This include is reserved for user modifications           *
* Forms for sales document processing                       *
* The name of modification modules should begin with 'ZZ’.  *
* * * * * * * * * * * * * * * * * * * * * * * * * * * * * * *
*
*&---------------------------------------------------------------*
*& Form USEREXIT_FILL_VBAP_FROM_HVBAP
*&---------------------------------------------------------------*
*       This user exit can be used to fill additional data into    *
*  VBAP from the main item (HVBAP), i.e. this Userexit is called   *
*       when an item is entered with reference to a main item.     *
*       This form is called from form VBAP_FUELLEN_HVBAP.          *
*---------------------------------------------------------------*
FORM userexit_fill_vbap_from_hvbap.
* VBAP-zzfield = HVBAP-zzfield2.
ENDFORM.
```


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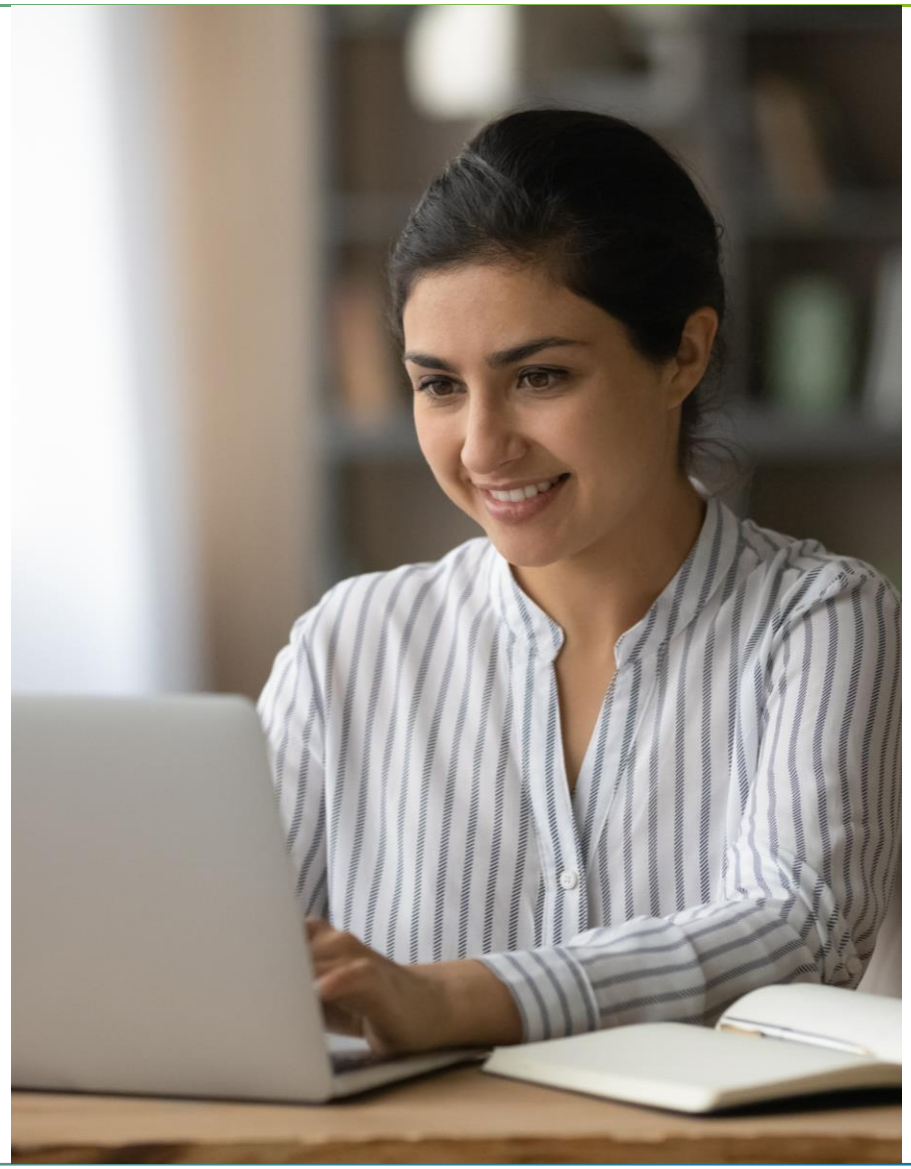
Customer Exits



Customer Exits Overview

- Created for specific programs, screens, and menus within standard applications
- Act as hooks and don’t contain any functionality. You can hang your own add-on functionality onto these hooks
- Are an `include` in the customer namespace and start with `CALL CUSTOMER`
- Types of Customer Exits:
  - Function Module exits
  - Screen exits
  - Menu exits

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Customer Exits

Advantages:

- Don’t affect standard SAP source code
- Don’t affect software updates

Disadvantages:

- Not available for all programs and screens found in the SAP System
- Can be used only if they already exist in the SAP System

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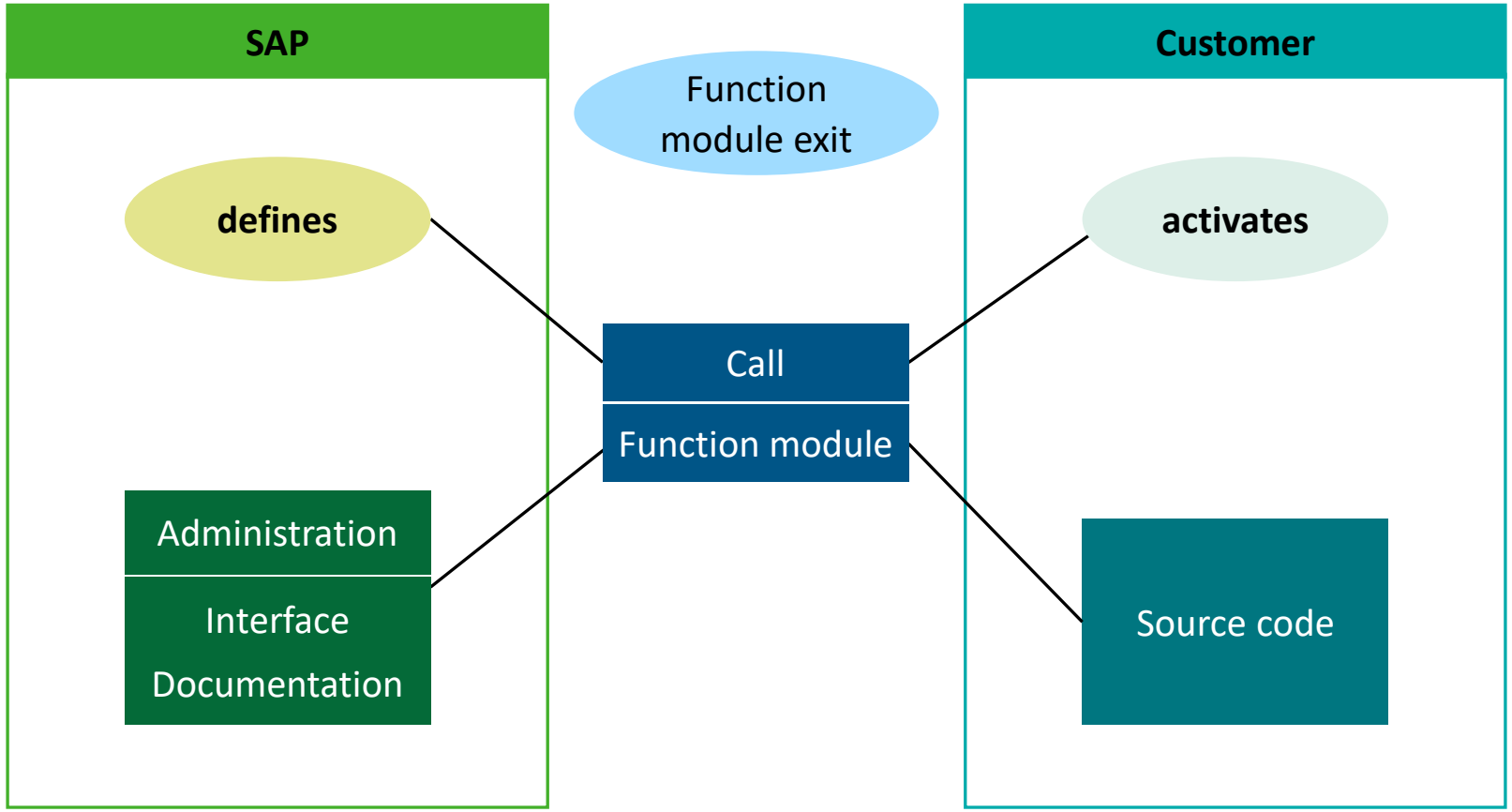
Program Exits Overview



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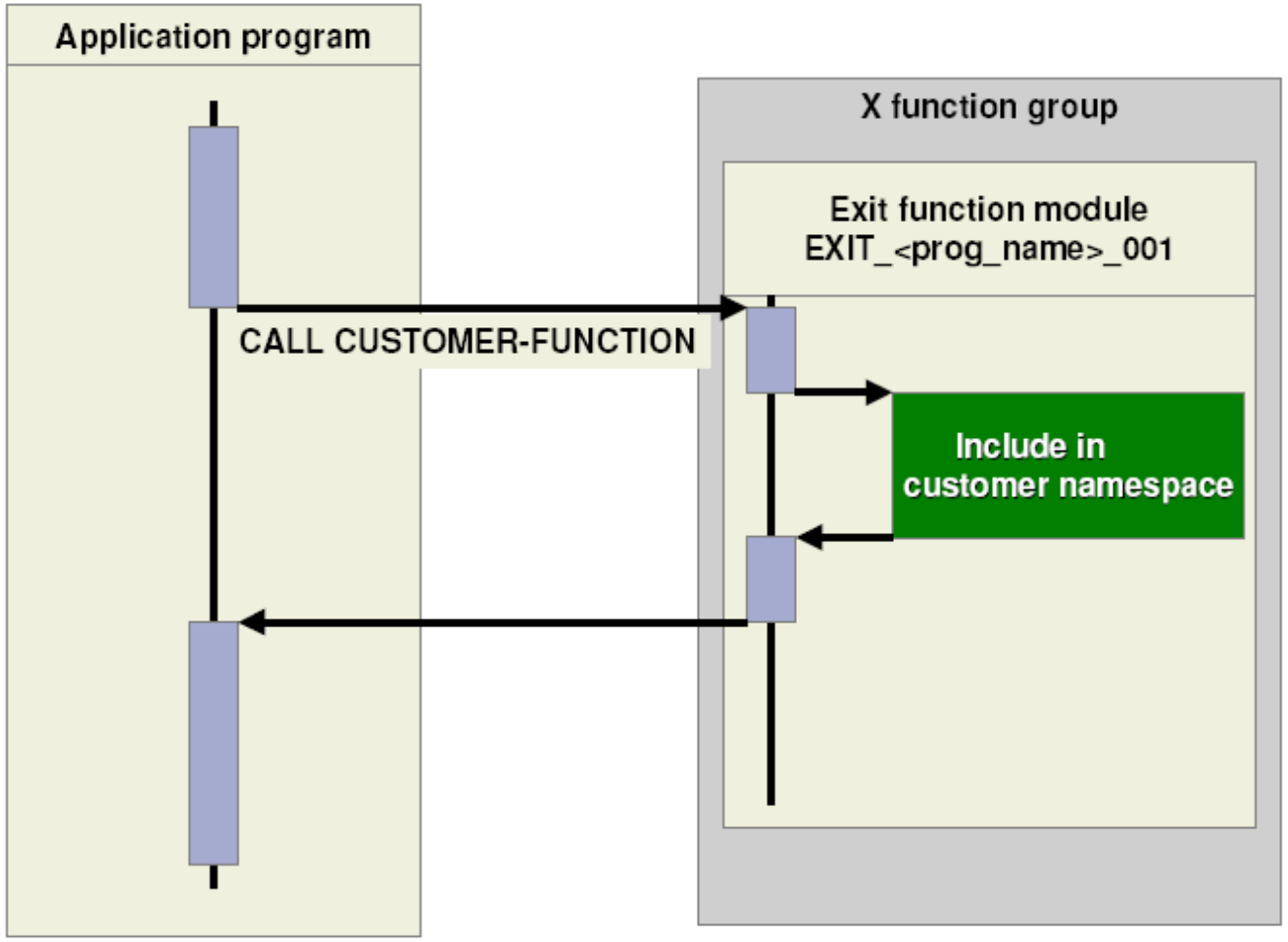
Function Module Exit—Process Flow



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Calling and Creating Function Modules

```
PROGRAM  <program_name>.

START-OF-SELECTION.
...

CALL CUSTOMER-
  FUNCTION '001'
        EXPORTING
           ....
        IMPORTING
           ....  .
```

```
FUNCTION-POOL XAAA.
...
FUNCTION exit_<program_name>.
...

  INCLUDE zxaaau01.



ENDFUNCTION.
```


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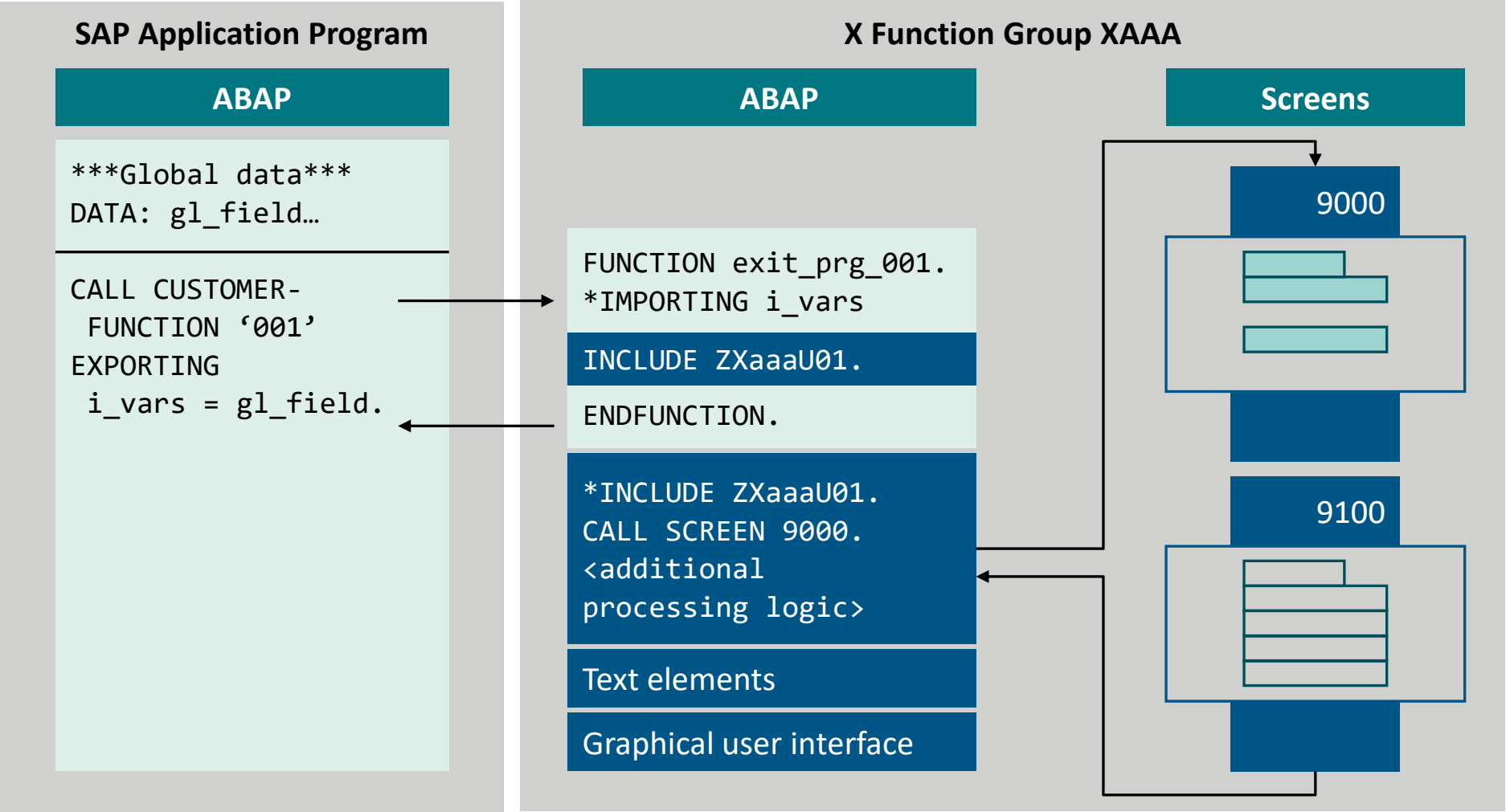
Summary—Function Module Exits



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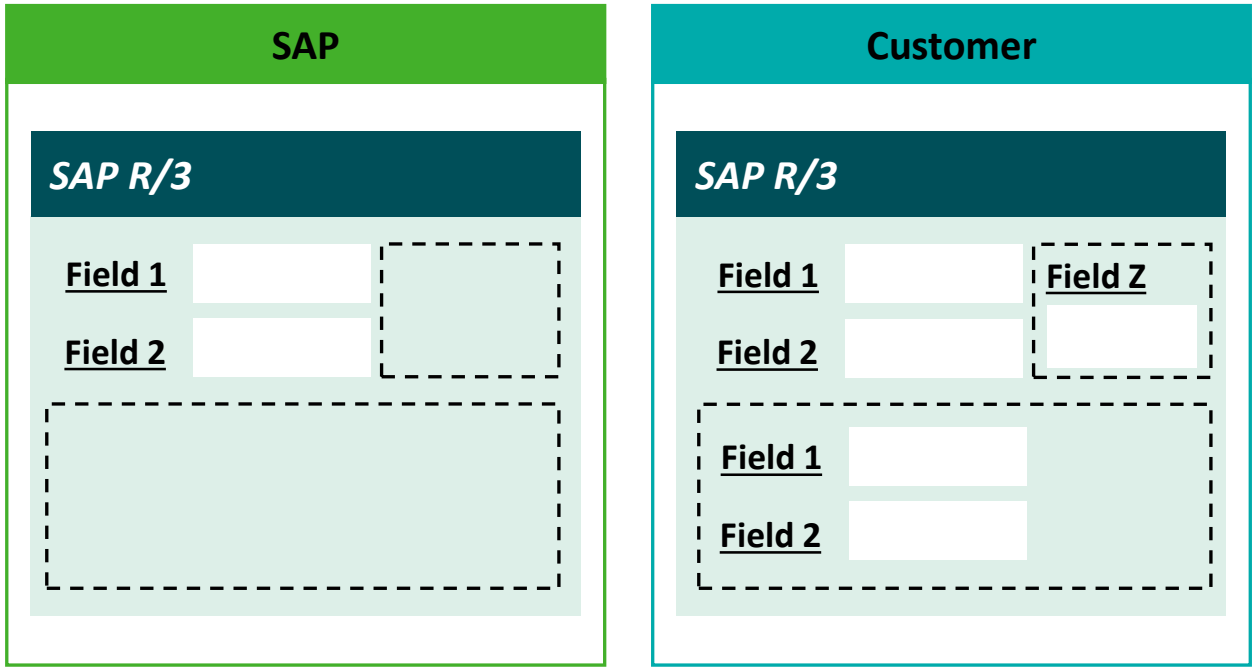
Screen Exits Overview

Screen exits belong to the customer exits. They allow displaying additional objects in an SAP application program screen. The SAP developer must:

- Define the subscreen areas
- Specify the corresponding calls in the flow logic
- Provide the framework for the data transport
- Include the screen exit in an enhancement



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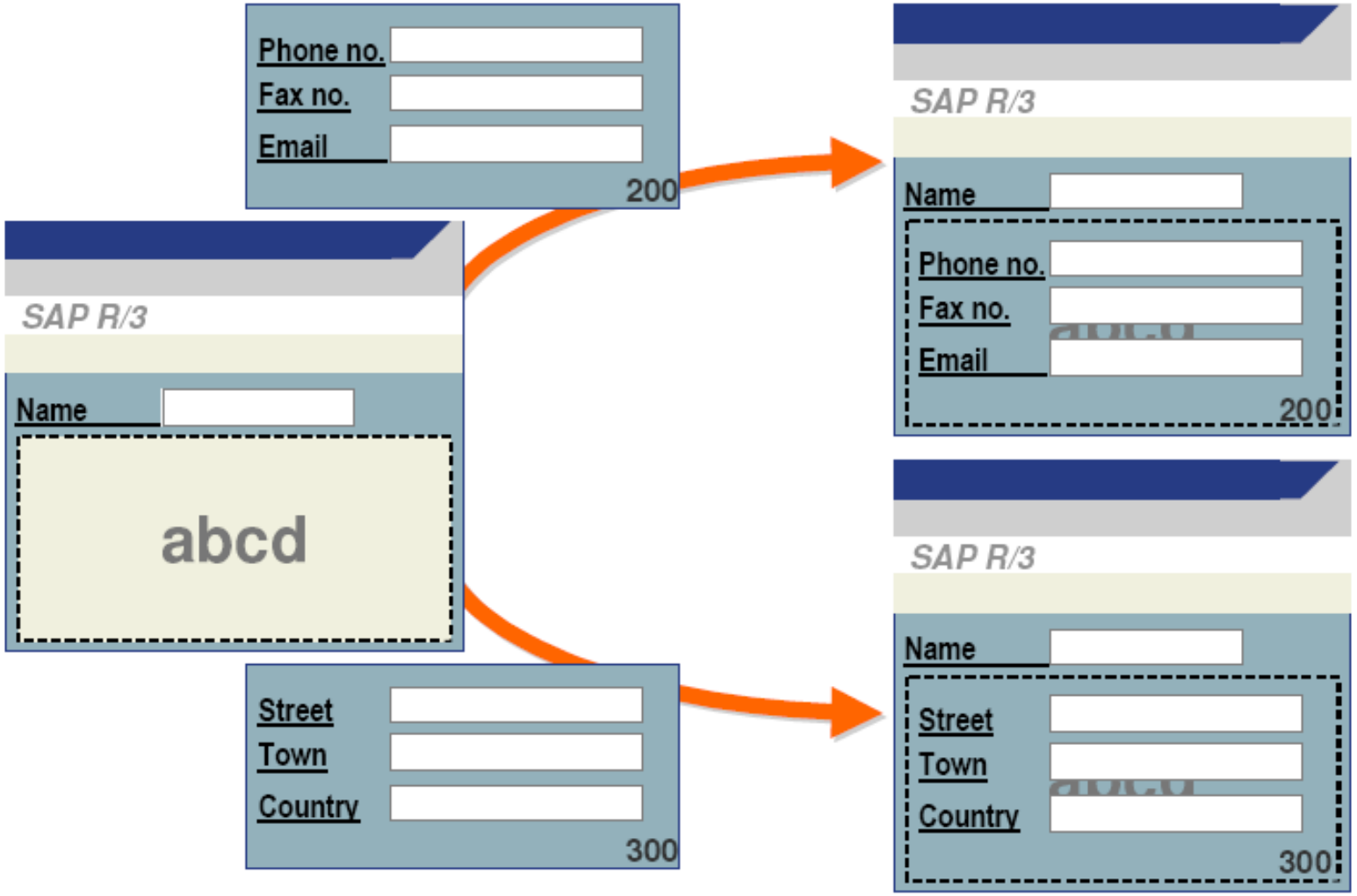
Subscreens in the R/3 System



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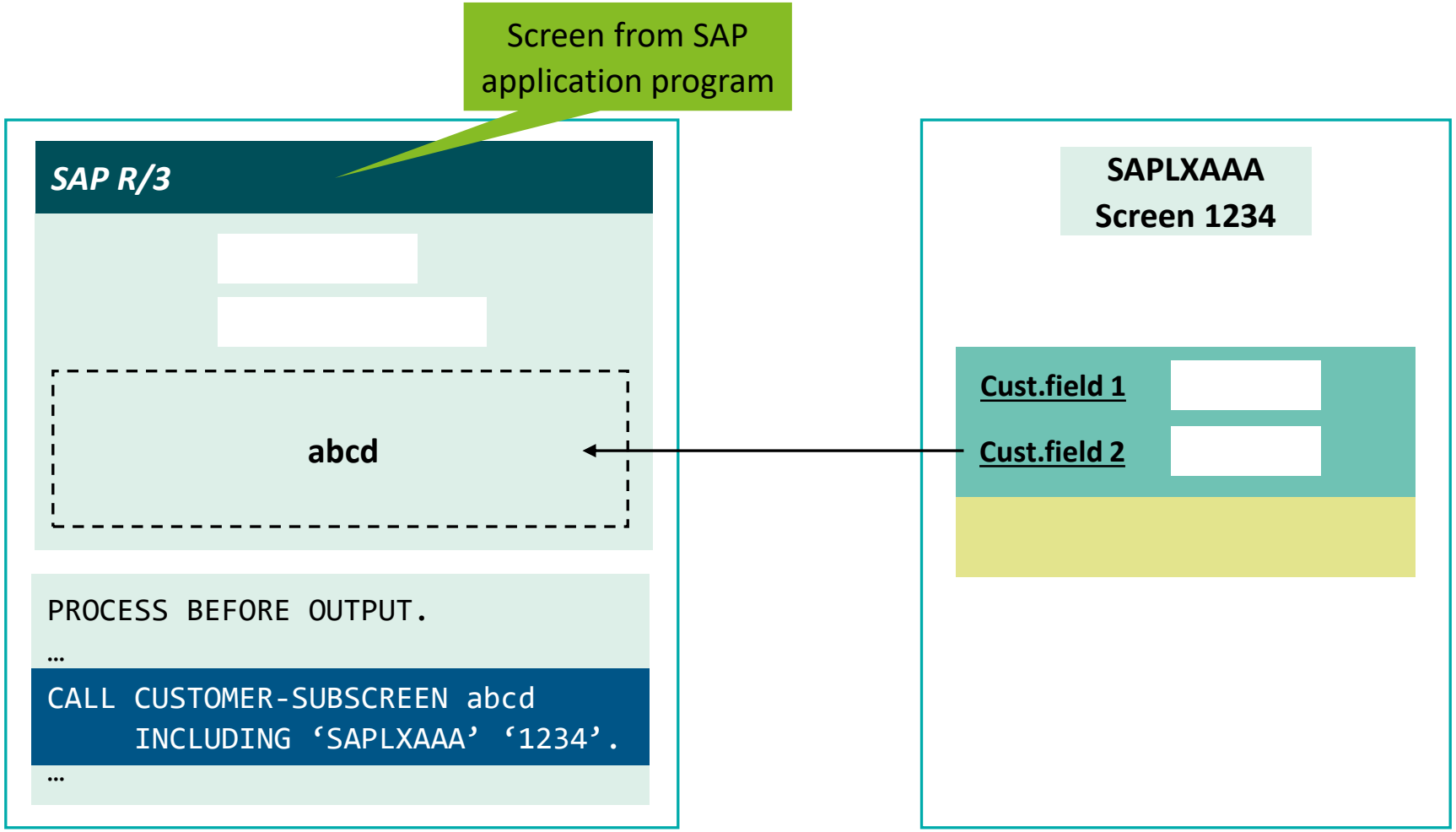
Defining Screen Exits



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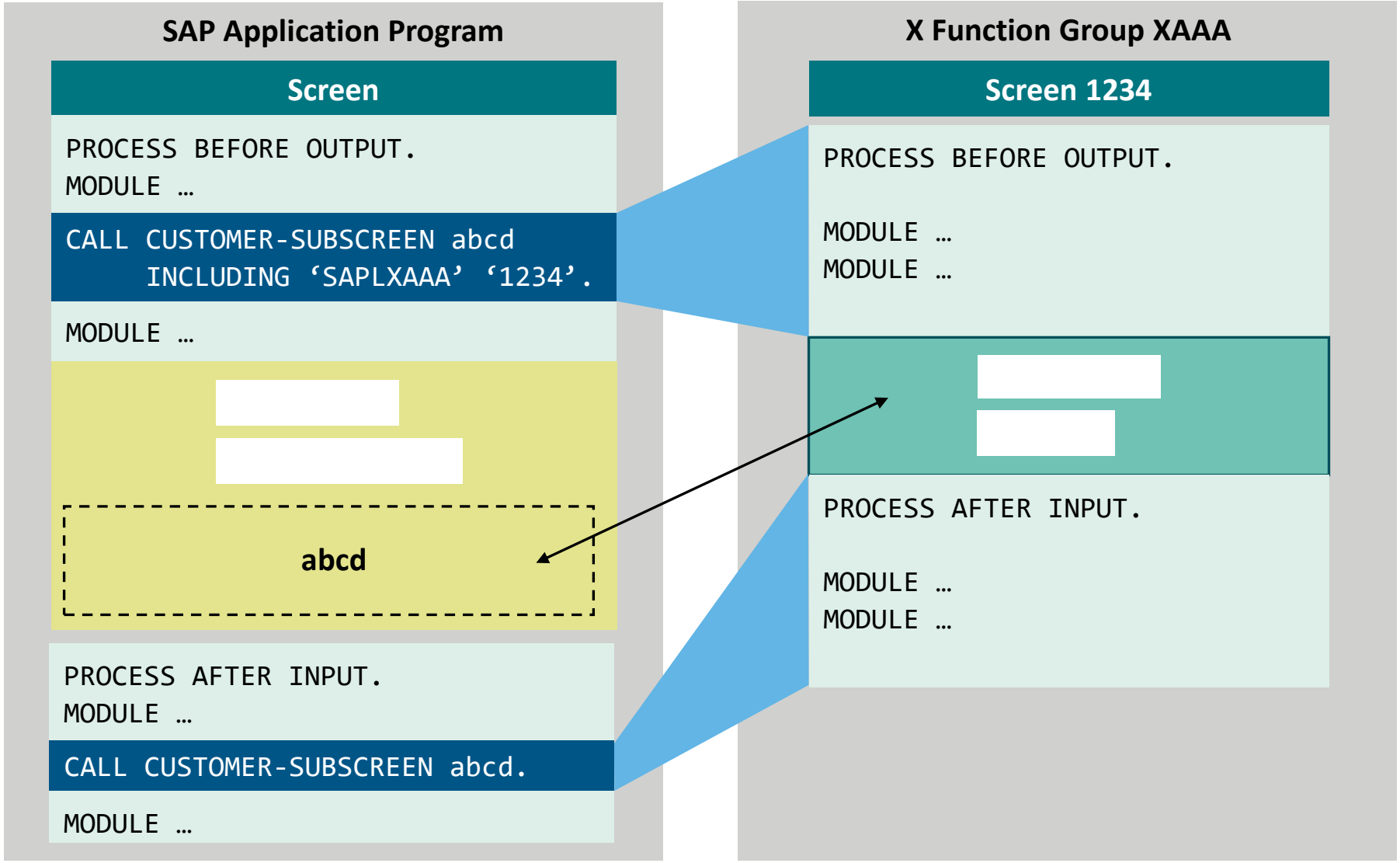
Calling Customer Subscreens



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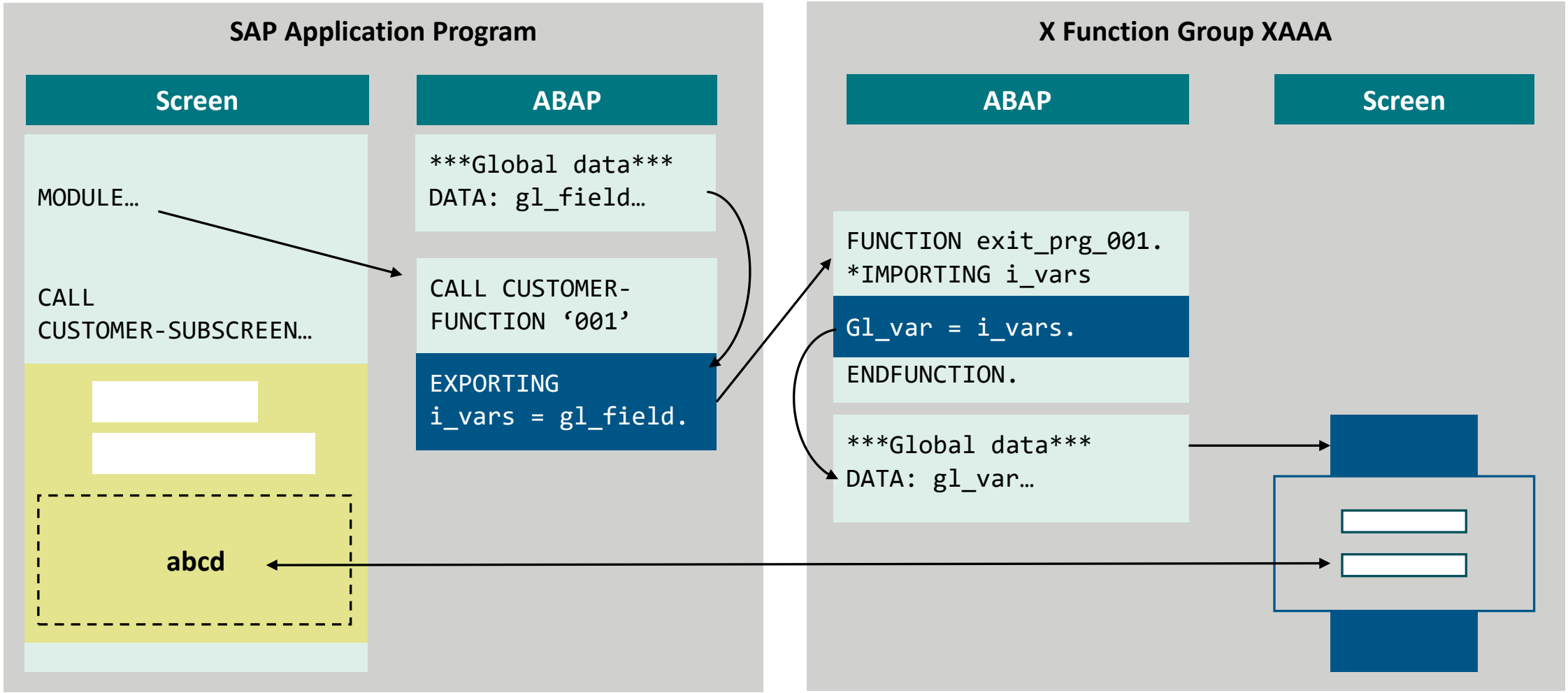
Transporting Data to Subscreens



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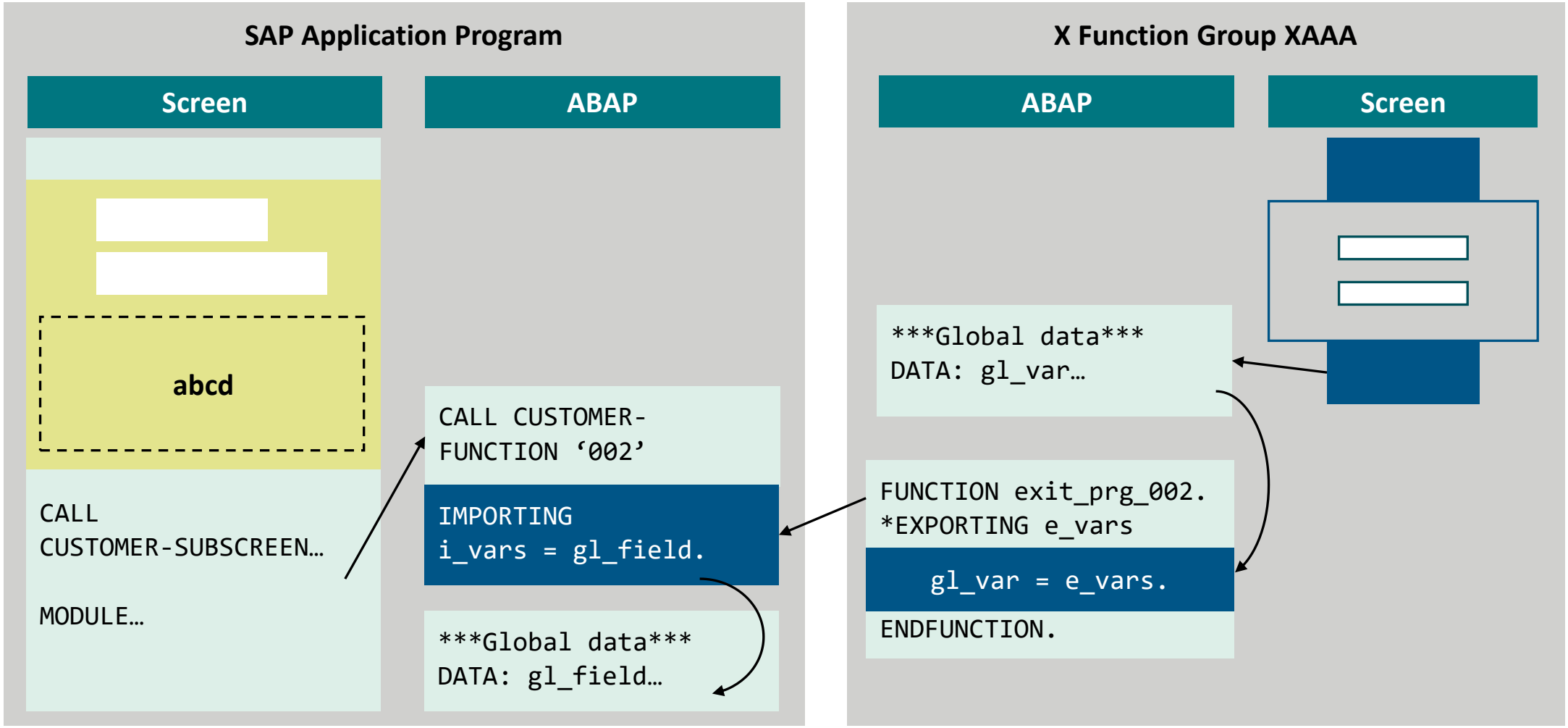
Transporting Data From Subscreens



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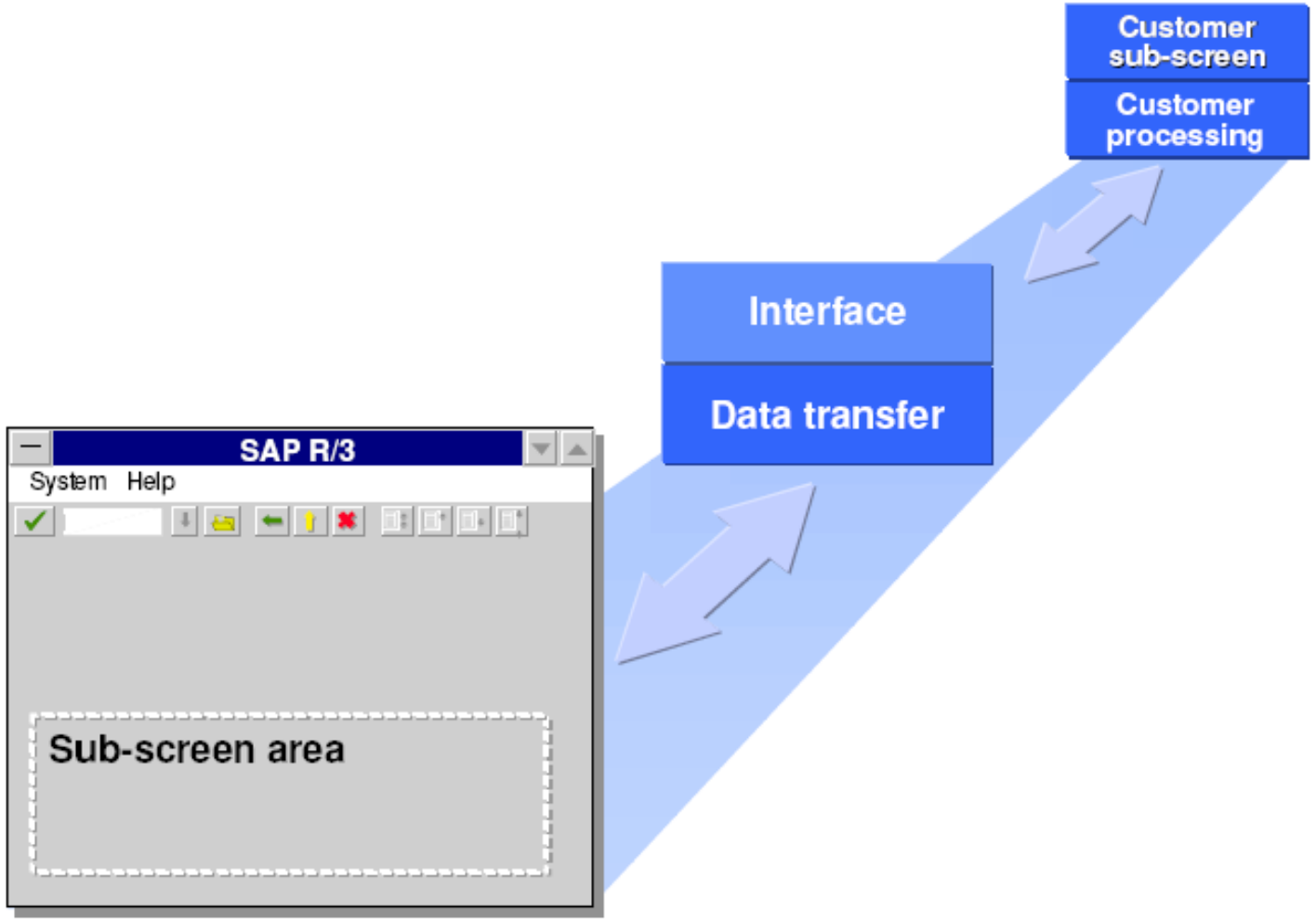
Summary—Screen Exits



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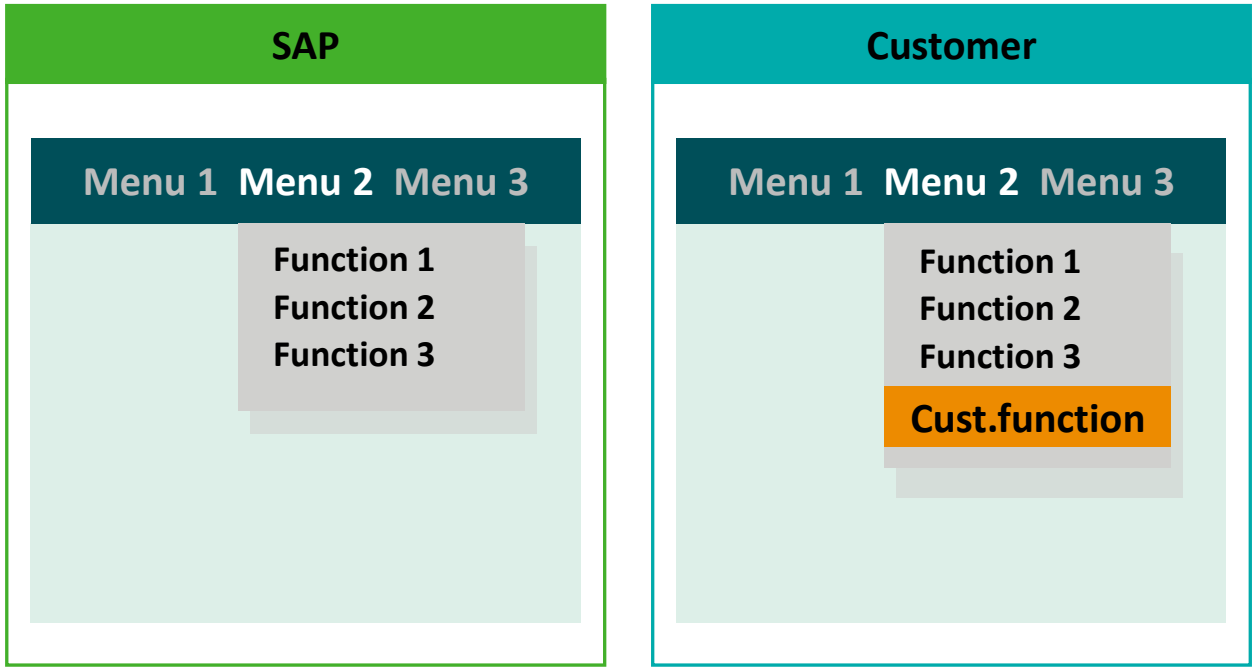
Menu Exits Overview

- Menu enhancements permit you to add additional menu entries to an SAP standard menu. The system provides two options here:
  - Customer exits
  - Business add-ins
- The additional menu entries are merged into the GUI interface
- When the function code is implemented, you can change the text of the menu entry, and—provided the SAP developer specified this option—change the icons



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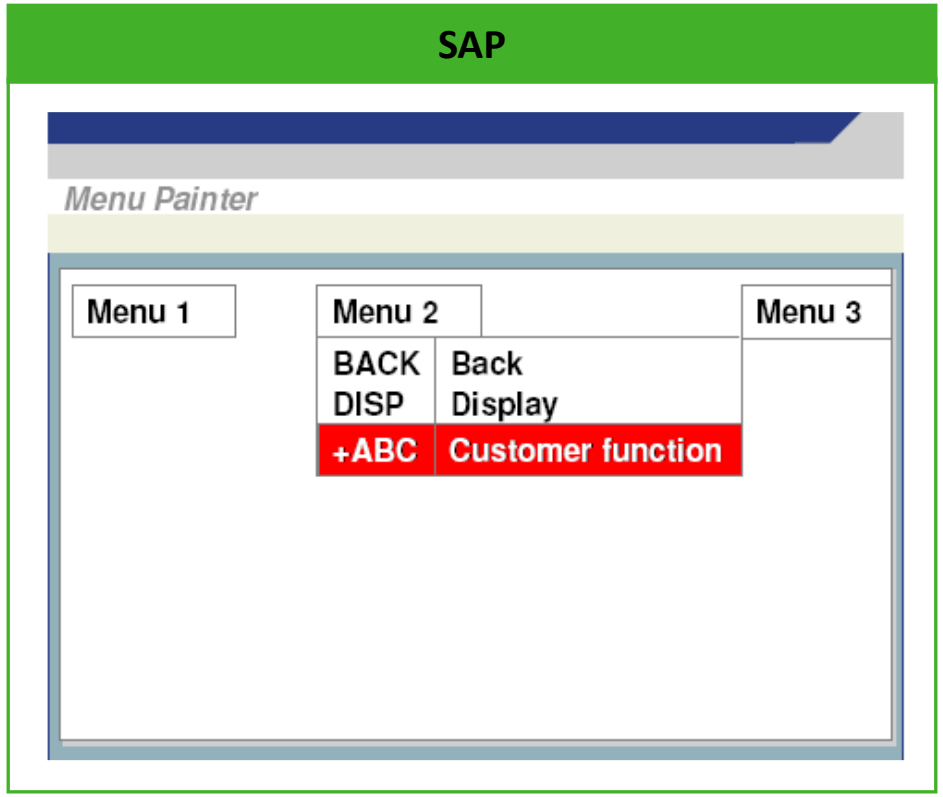
Menu Exit Requirements



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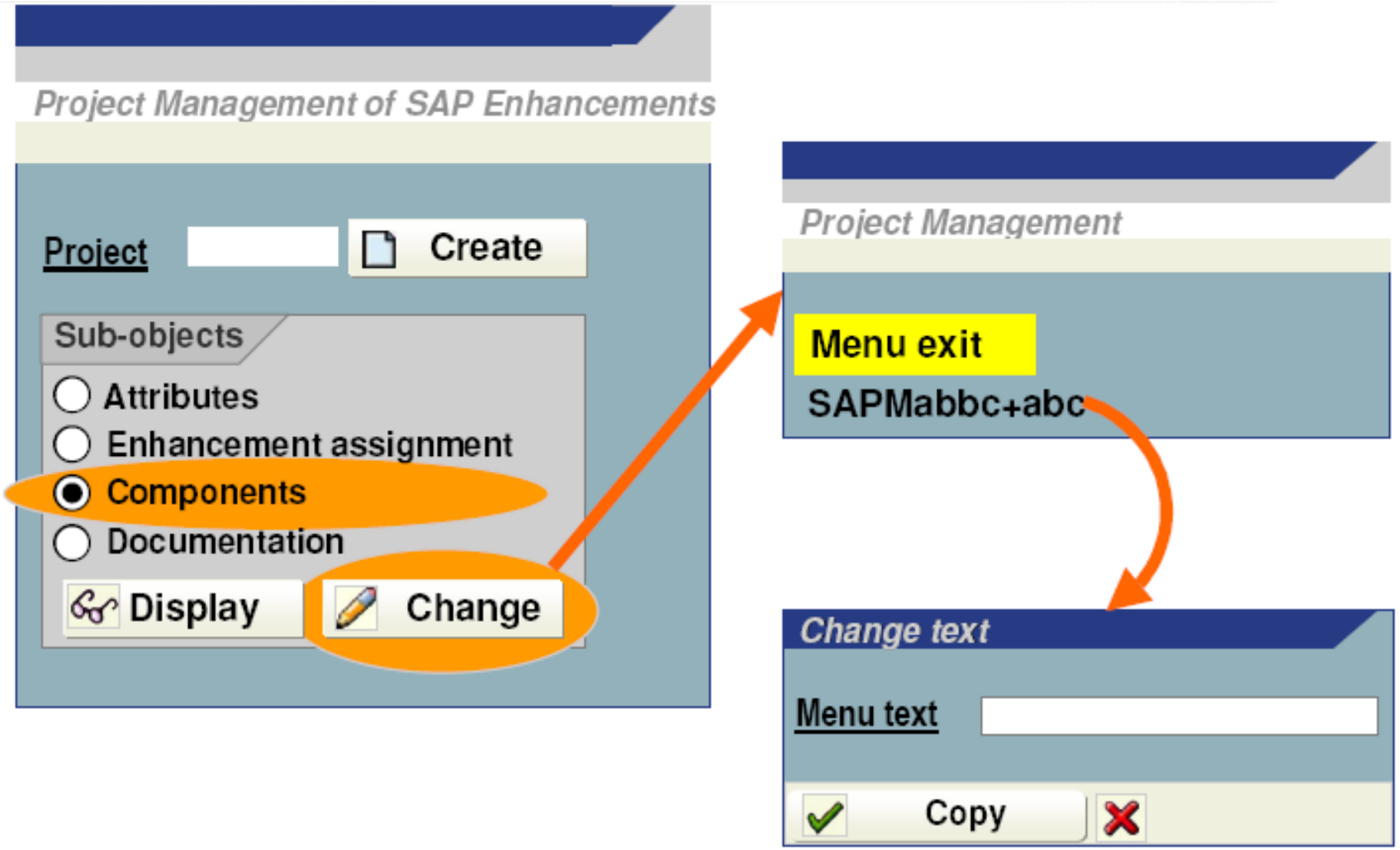
Editing Menu Exits



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Menu Exits and Function Module Exits

```
PROGRAM  <program_name>.
DATA     ok_code  LIKE sy-ucomm.

...

CASE ok_code.
  WHEN 'DISP'.
    ...
  WHEN '+ABC'.
    CALL CUSTOMER-FUNCTION '001'
           EXPORTING
             <i_variables>
           IMPORTING
             <e_variables>.

    ...
ENDCASE.
```


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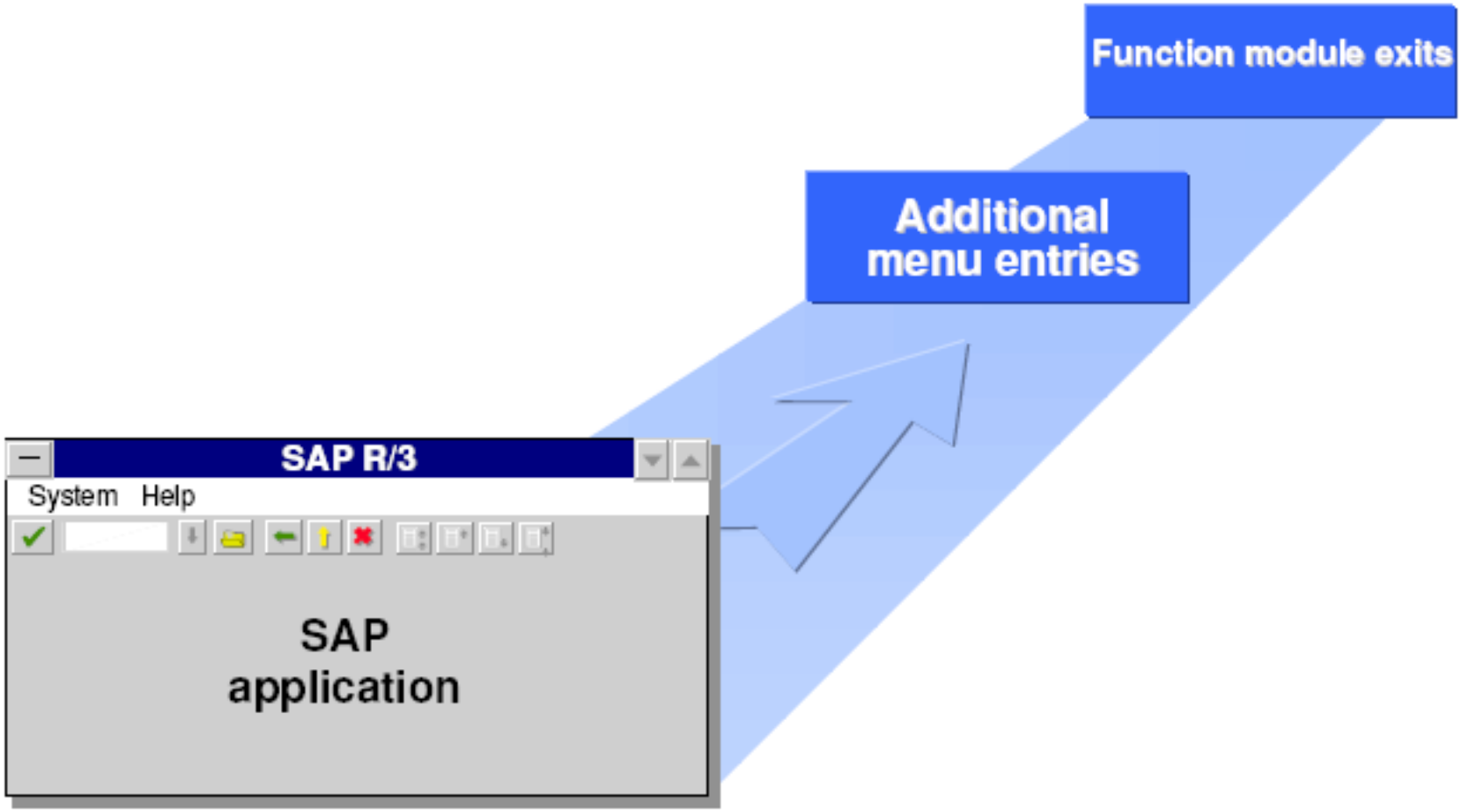
Summary—Menu Exits



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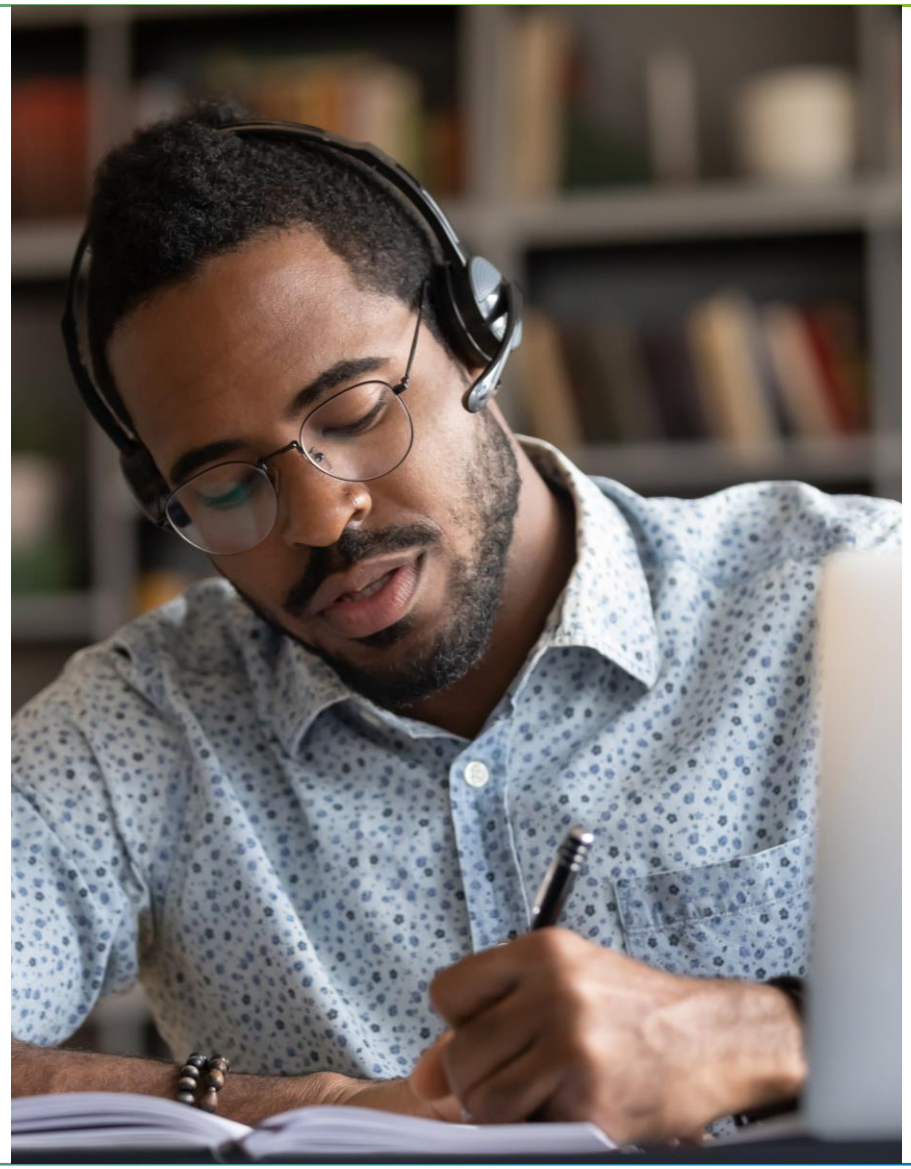
Hands-On Labs



Instructions

- Project Naming Convention: ZCR15A<Sequence Number>
- Include Naming Convention: ZITRAINEE<ID>

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Hands-On Activity 1

Activity Details: User Exit Exercise	
Problem Statement	Identify the User exit during sales order save. Store sales order, item number, user name, date, and time creation in a custom table.

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Hands-On Activity 2

Activity Details: Customer Exit Exercise	
Problem Statement	Identify the custom enhancement which triggers the SAVE function during the creation of customers from standard SAP transactions (XD01). On Save validate if the country entered in the transaction is ‘US’. If it is not the US then gives an error message “Create customer with country code US — <Trainee ID>”.

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Hands-On Activity 3

Activity Details: Screen Exit Exercise	
Problem Statement	Enhance IE01/02/03. (Refer to the list below for the tcodes). a. Create your own custom fields (1 character, 1 numeric, and 1 alphanumeric) and store the value in a custom table. b. Custom table naming (ZZ<USERID>_EQ) c. Custom field naming (ZZ_<FIELDNAME>) d. Validation should be there for numeric and character fields. e. Fields should be input disabled in display transaction.

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Hands-On Activity 4

Activity Details: Menu Exit Exercise	
Problem Statement	Enhance the MC94 Menu bar and add a new user command to Goto/Utilities in the Menu bar. Name the new function text as ‘On click’. After the function ‘On Click’ has been added to Utilities, when the user clicks on it (Utilities -> On Click) a message should be displayed saying ‘Function ‘On Click - <User ID>’ has been triggered’. Planning type: Z_S140
Steps	Enhancement Name: Find it out Create a project using transaction code CMOD and assign the enhancement. Add the new function and activate the project. Check for the function exit and write the code when this function is clicked. Display an information message when this function is clicked.

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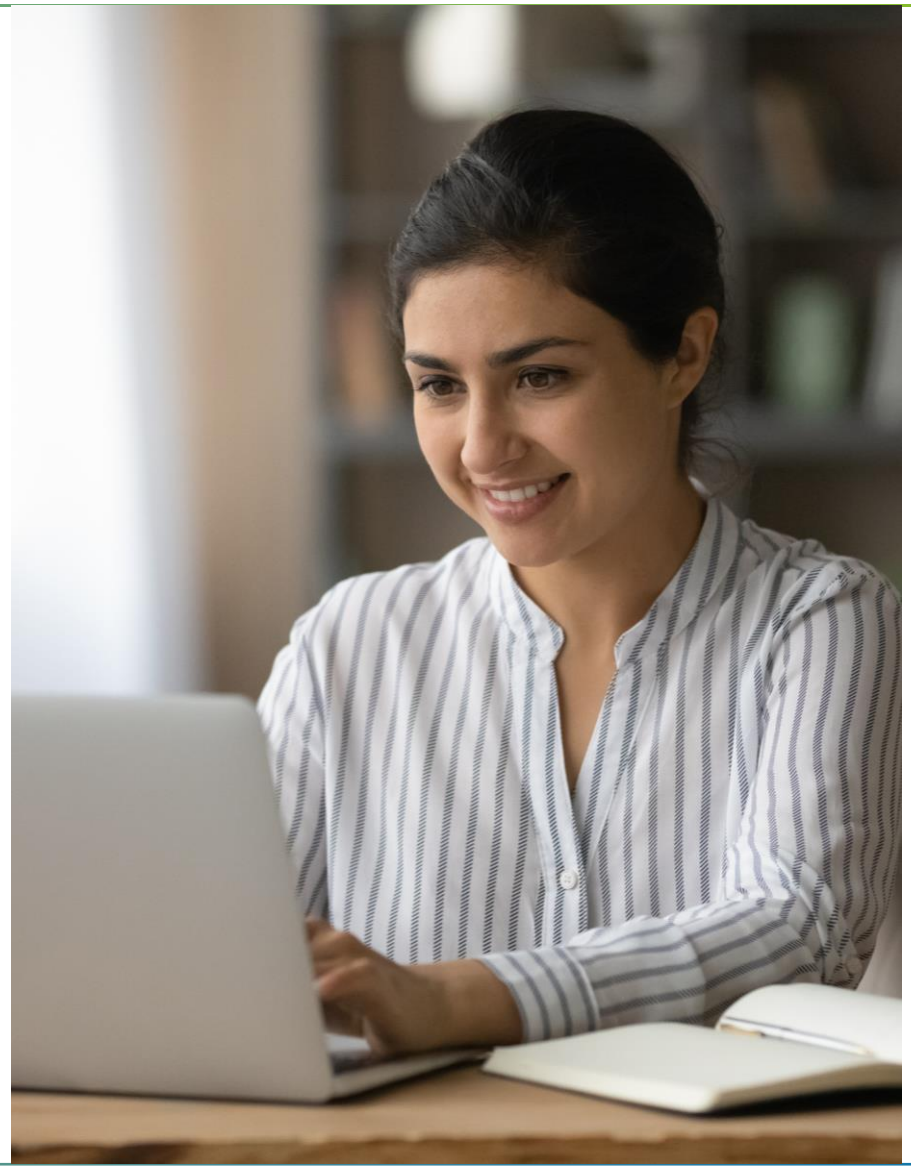
Business Add-Ins (BADI)



BADI Components

BADI

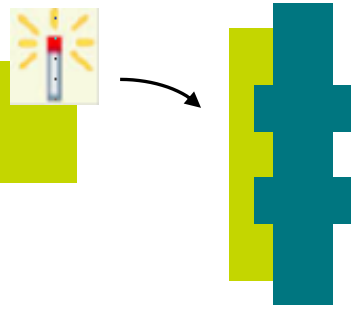
Attributes

- [x] Reusable
- [ ] Filter-dependent

FCodes

Program	Function code
<prog>	+ABC

```
INTERFACE IF_EX_<badi>.
DATA:        a1 …
METHODS m1
  EXPORTING
    e_par1
    e_par2
  IMPORTING
    i_par1.
METHODS m2 …
END INTERFACE.
```



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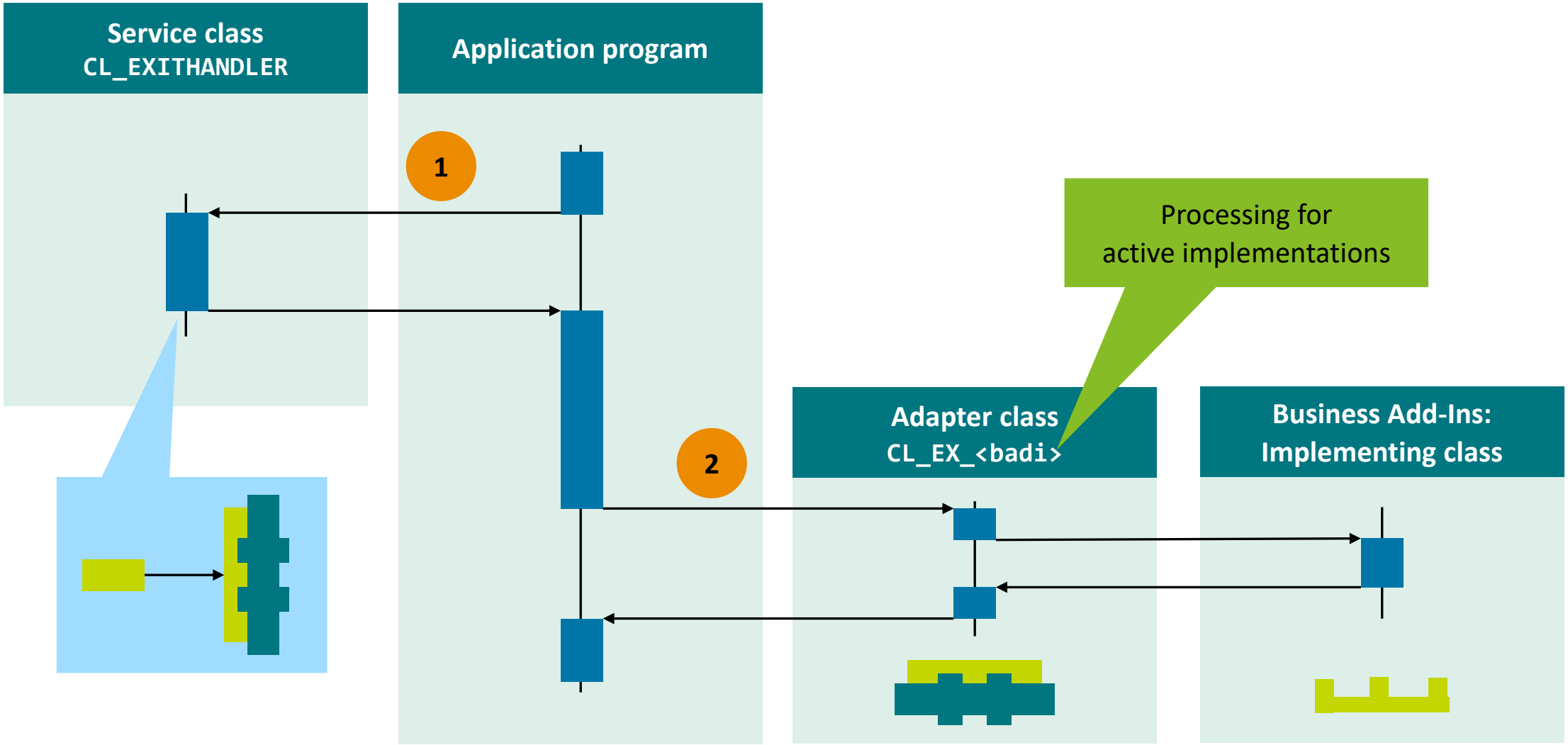
BADI Process Flow Diagram



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BADI: Calling Programs

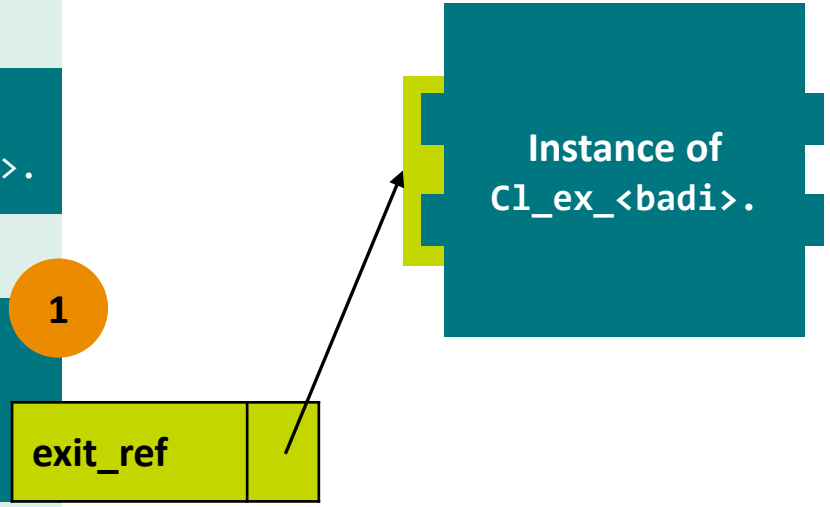
```
REPORT <program_using_badi>.

DATA: exit_ref
        TYPE REF TO IF_EX_<badi>.

START-OF-SELECTION.
CALL METHOD
  cl_exithandle=>get_instance
    CHANGING
        instance = exit_ref.

CALL METHOD exit_ref->method
       EXPORTING
         <i_variables>
       IMPORTING
         <e_variables>.

END-OF-SELECTION.
```



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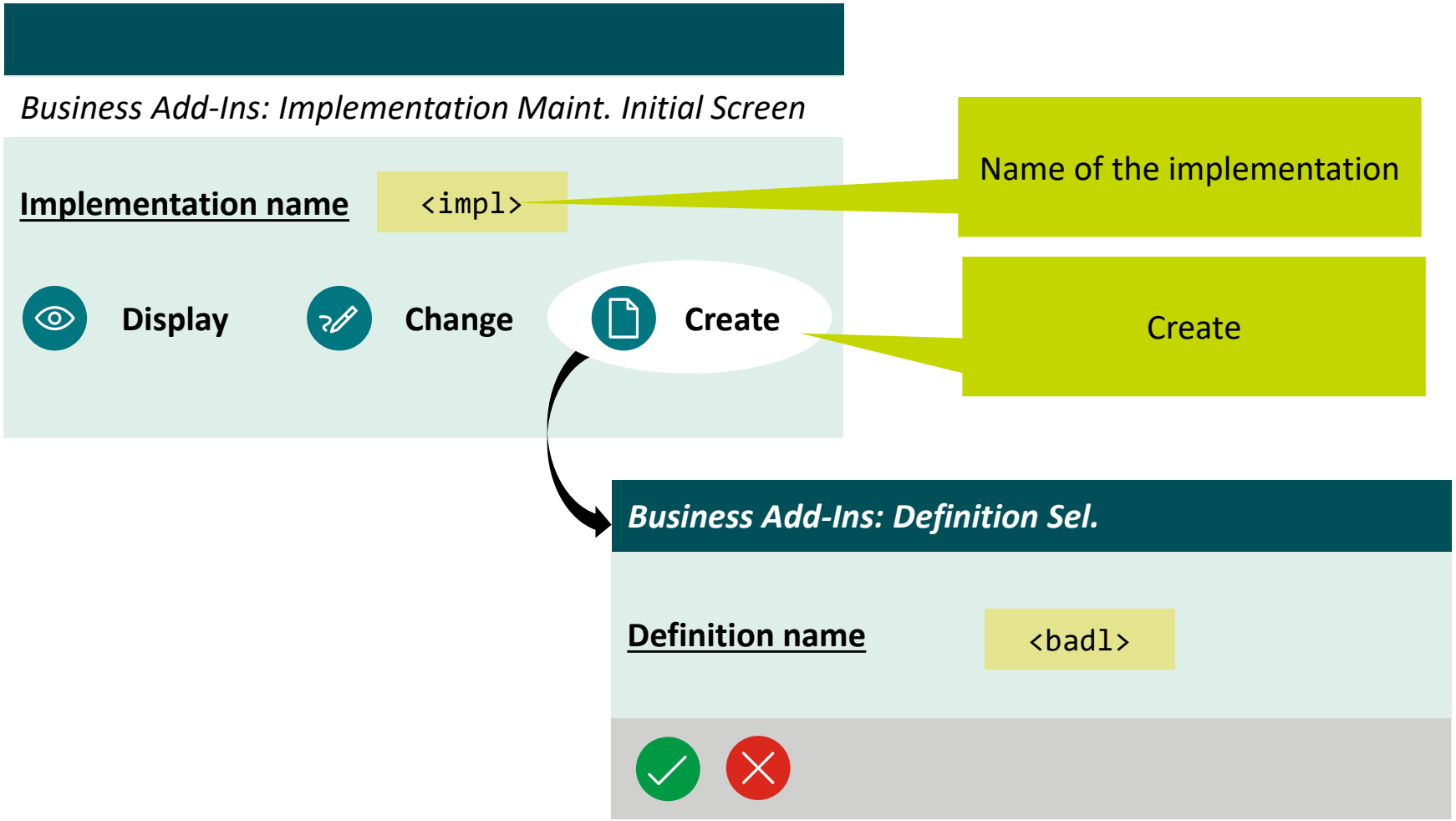
Implementing BADI—Initial Screen



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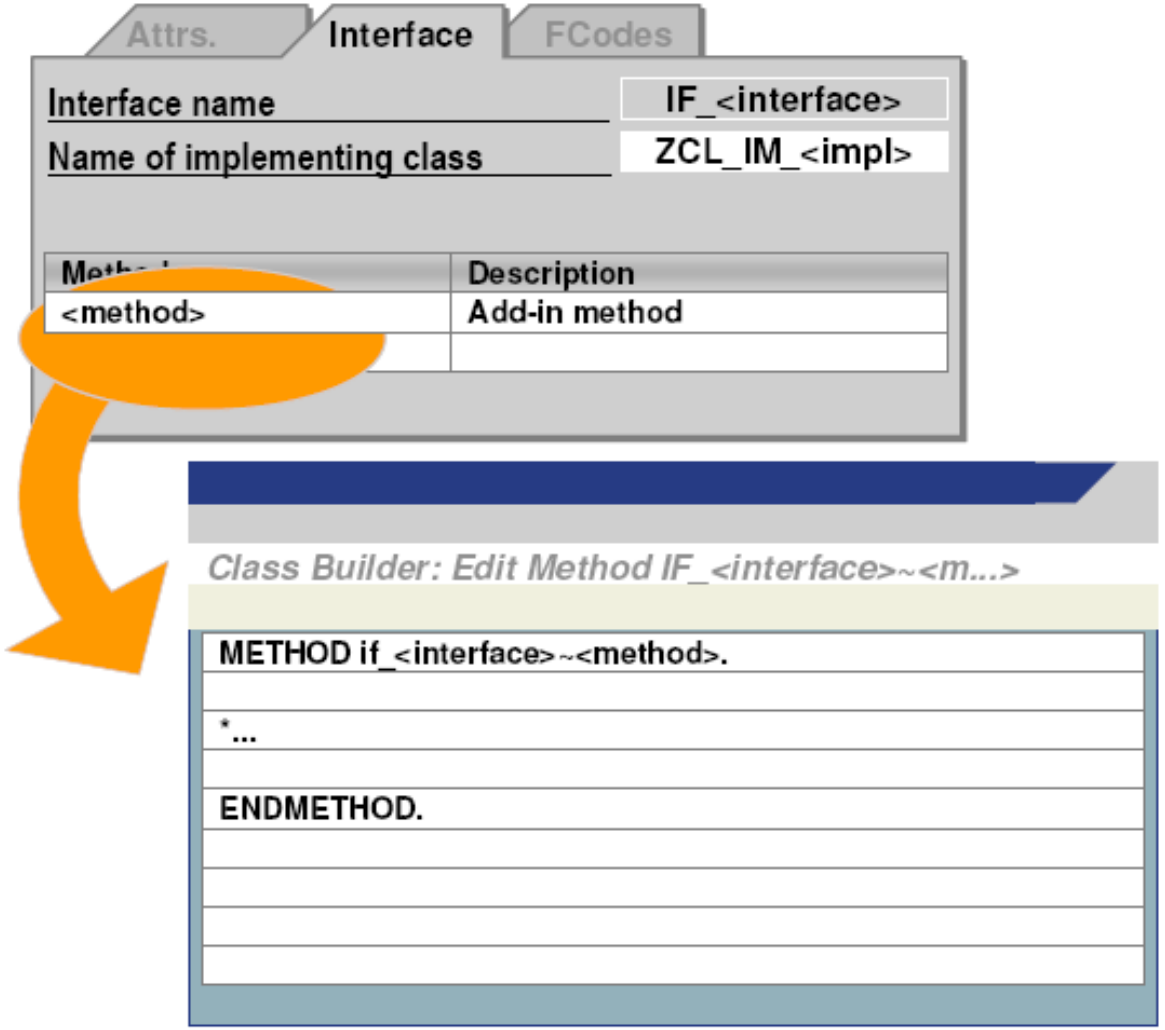
Implementing BADI—Methods



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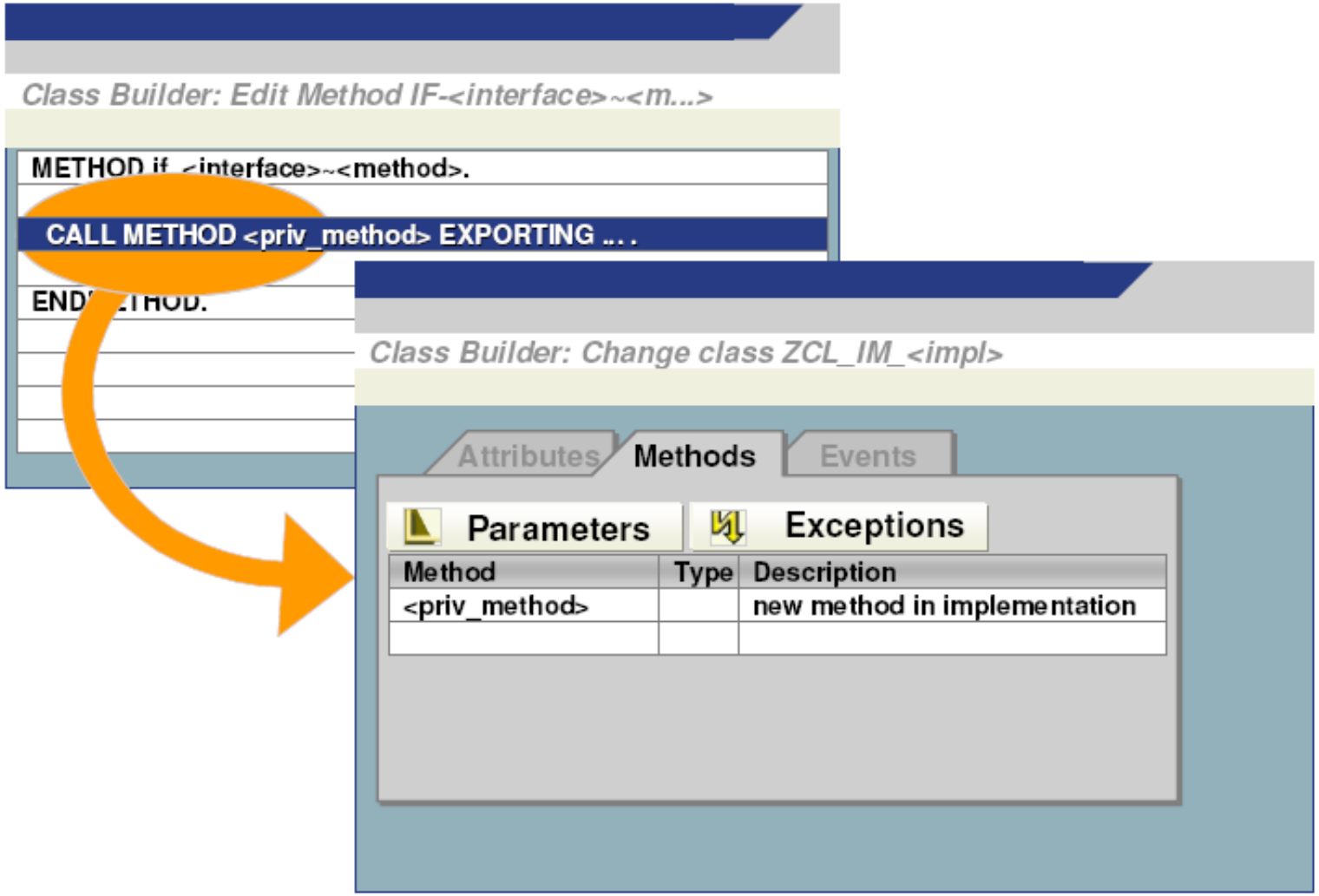
Implementing BADI—Private Methods



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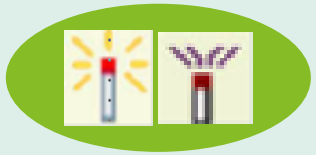
Implementing BADI—Activation

Business Add-Ins: Implementation Maint. Initial Screen



Implementation name `<impl>`

Display **Change** **Create**

- Activation
- Undo activation

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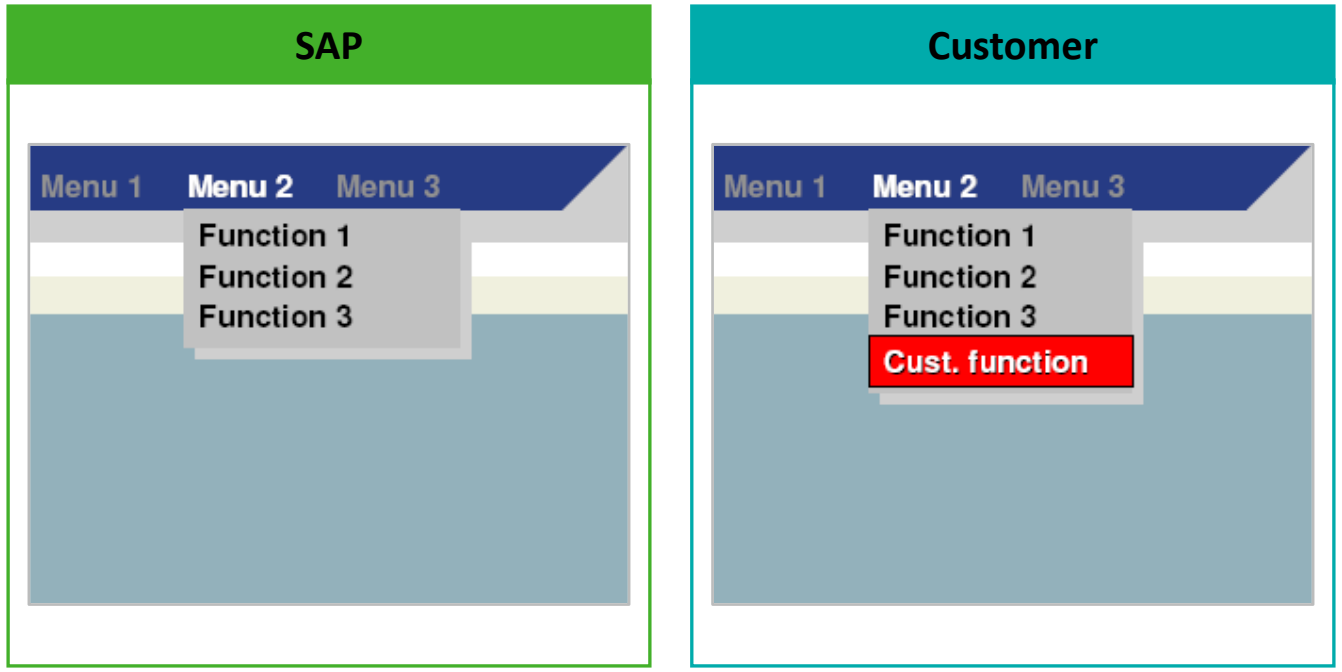
BADI Function Codes—Overview



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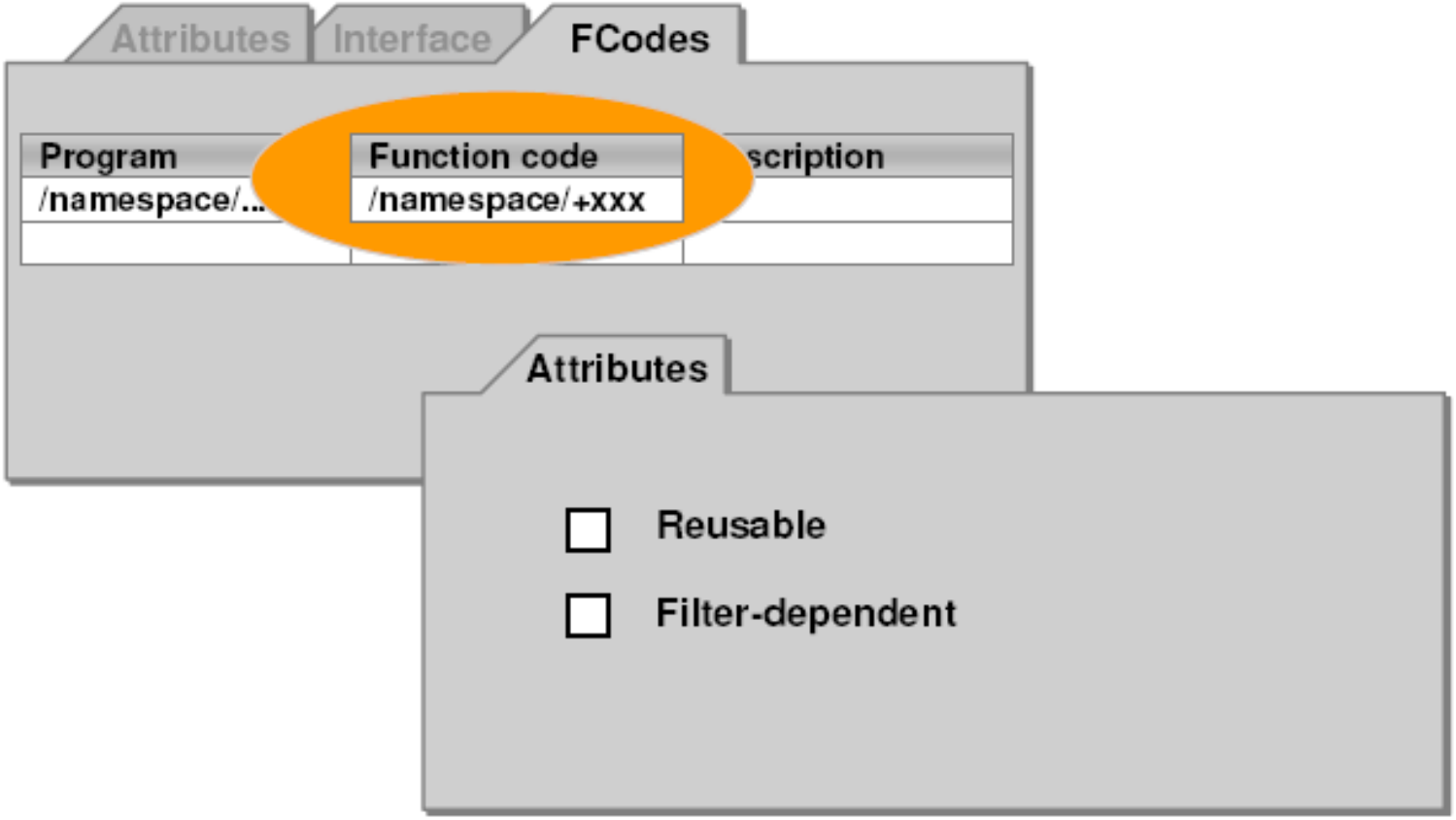
BADI Function Codes—Restrictions



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BADI Function Code Processing in the Program

```
REPORT <program_using_badi>.
DATA     ok_code LIKE sy-ucomm.
DATA:    exit_ref TYPE REF TO_IF_EX_<badi>.
…

CASE ok_code.
  WHEN 'DISP’.
  WHEN '+XXX’.
    CALL METHOD exit_ref->method
            EXPORTING
              <i_variables>
            IMPORTING
              <e_variables>.

ENDCASE.
```


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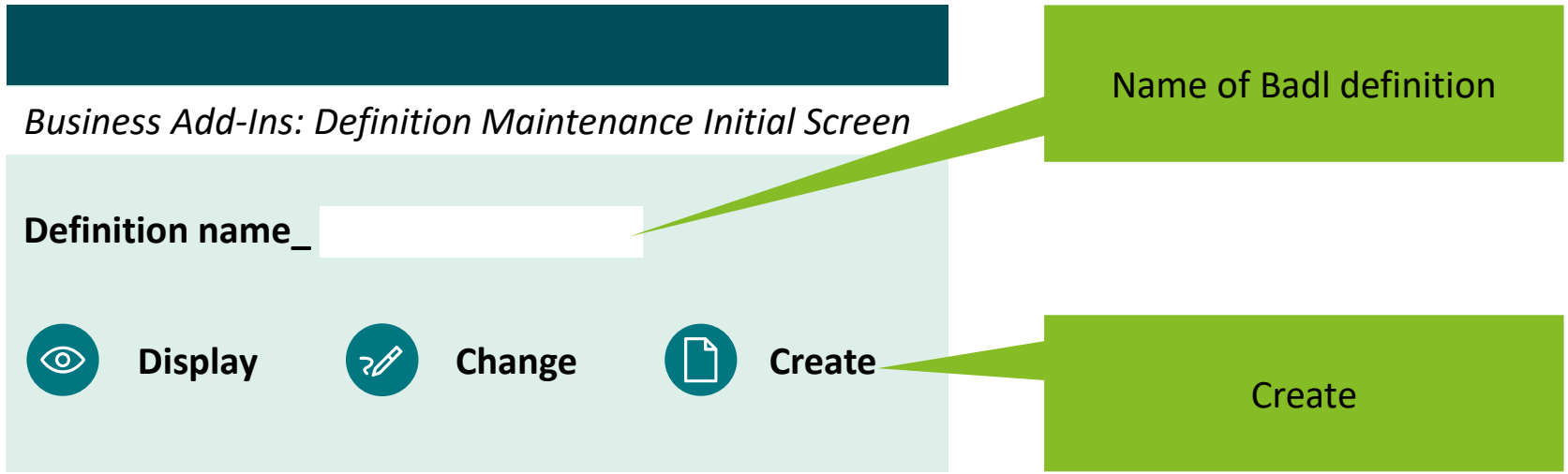
BADI Definition—Initial Screen



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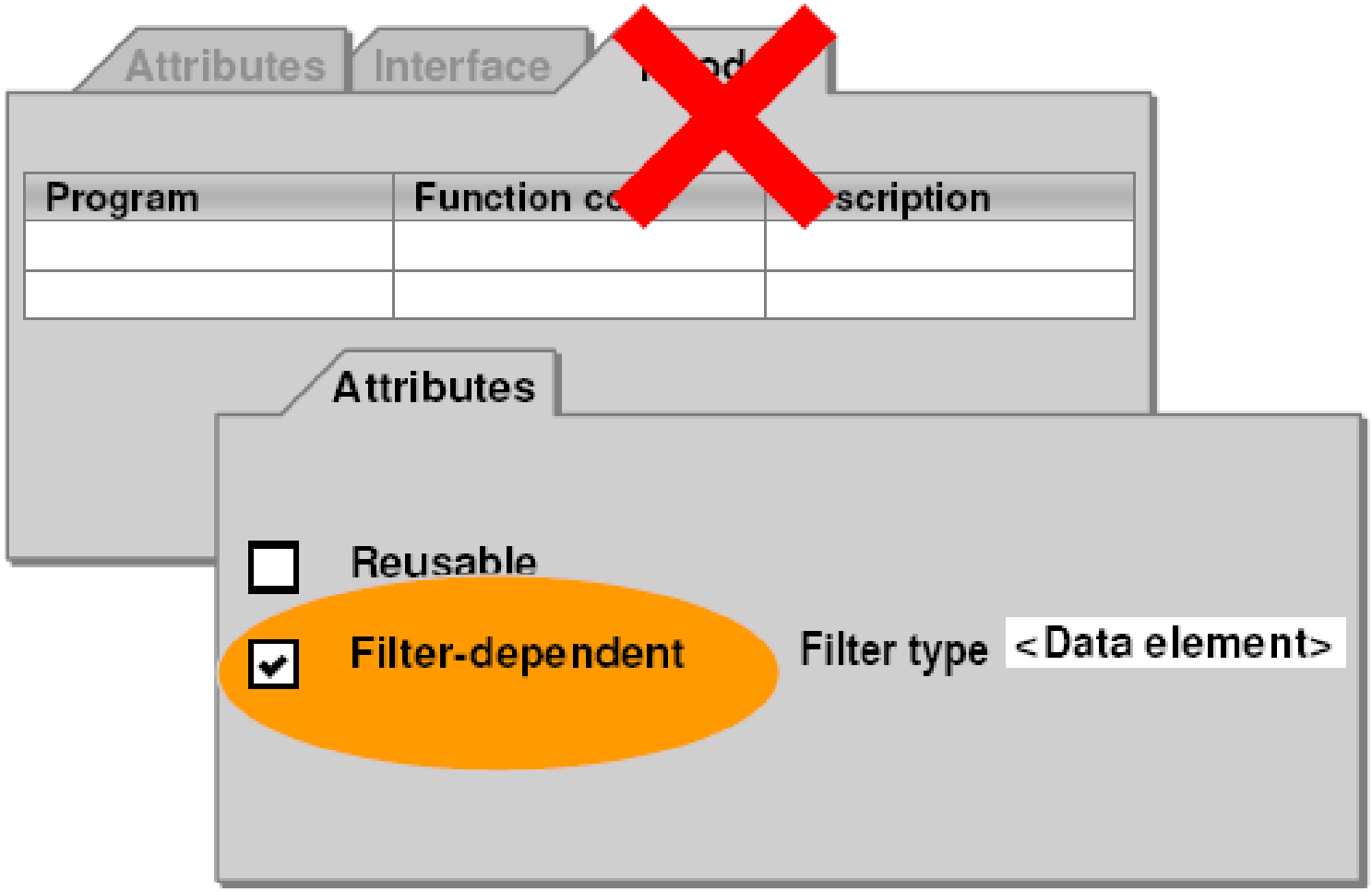
BADI Definition—Attributes



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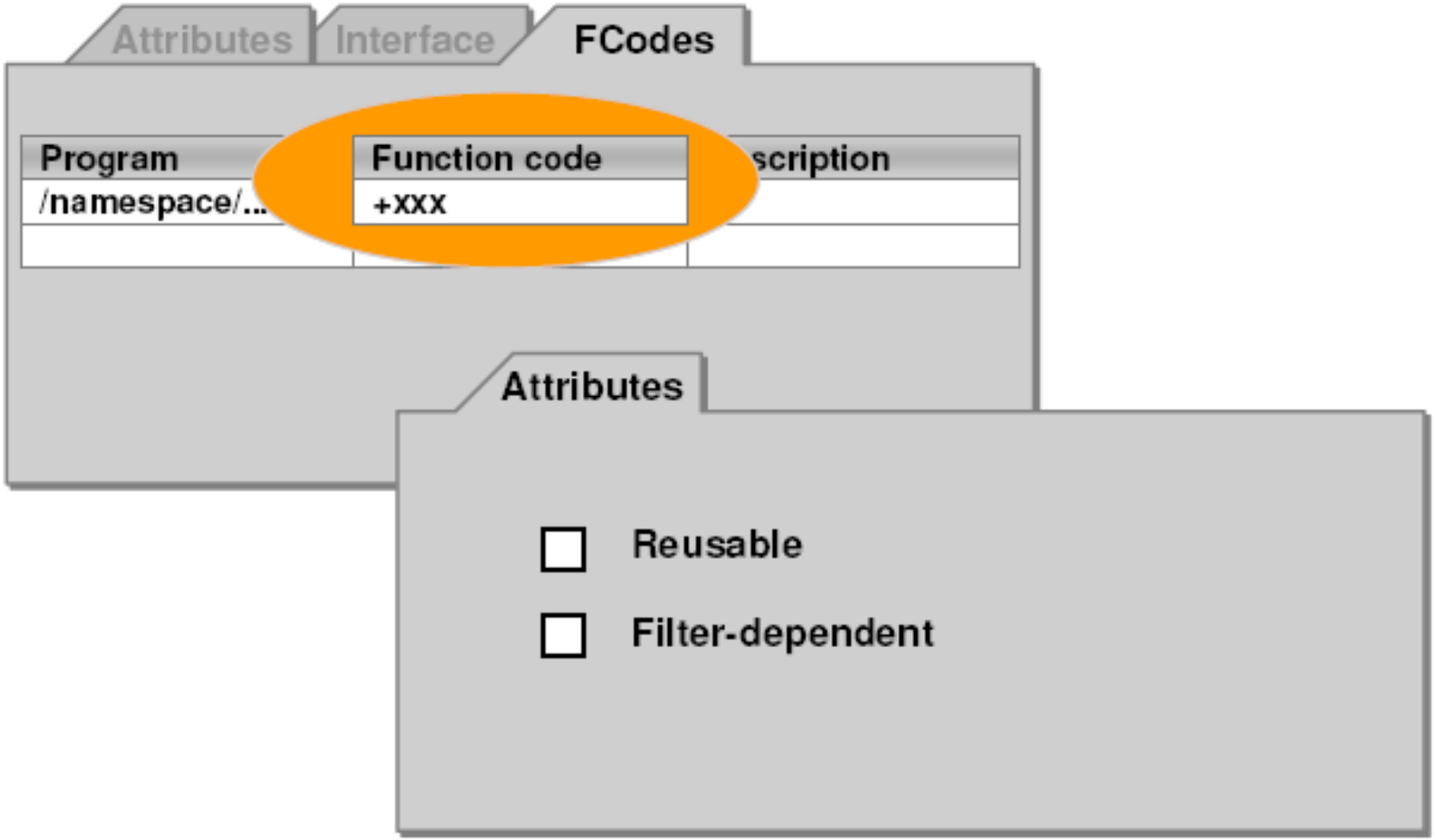
BADI Definition—Function Codes



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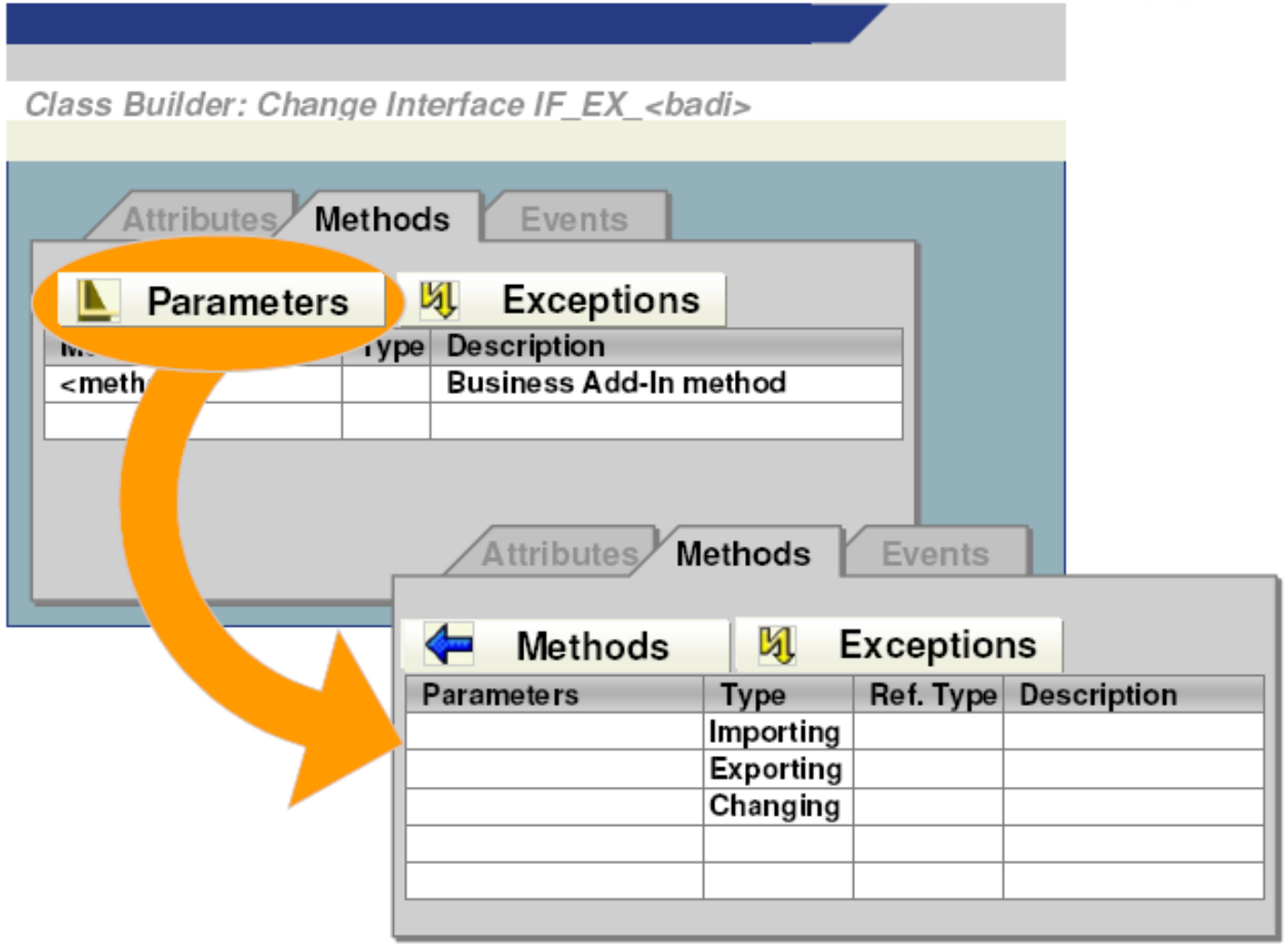
BADI Definition—Method Interface Parameters



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BADI Definition—Activating the Interface

Class Builder: Change Interface IF_EX_<badi>.

Activation

Attrs. | **Methods** | Events

Parameters **Exceptions**

Method	Type	Description
<method>		BAdI method

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BADI Definition—Call in Program

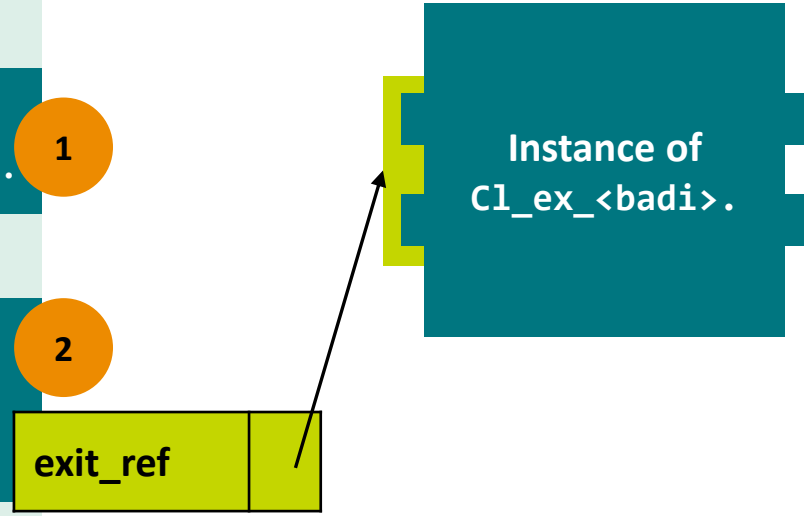
```
REPORT <program_using_badi>.

DATA: exit_ref
        TYPE REF TO IF_EX_<badi>.

START-OF-SELECTION.
CALL METHOD
  cl_exithandle=>get_instance
    CHANGING
        instance = exit_ref.

CALL METHOD exit_ref->method
       EXPORTING
         <i_variables>
       IMPORTING
         <e_variables>.

END-OF-SELECTION.
```



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Calling a Filter-Dependent Business Add-In

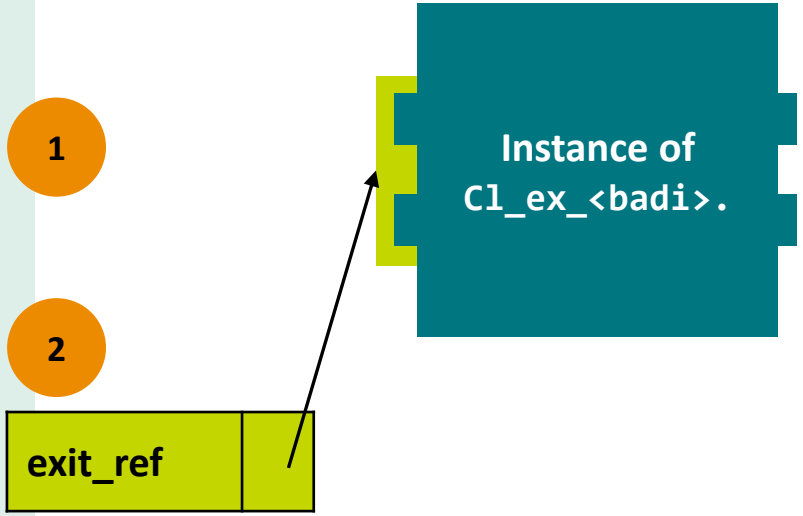
```
REPORT <program_using_badi>.
DATA: exit_ref
       TYPE REF TO IF_EX_<badi>.

START-OF-SELECTION.

CALL METHOD
  cl_exithandle=>get_instance
    CHANGING
        instance = exit_ref.

CALL METHOD exit_ref->method
       EXPORTING
         <i_variables>
       IMPORTING
         <e_variables>.

END-OF-SELECTION.
```



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BADI Definition: Naming Conventions

BADI Definition	`<badi>` or `Z<badi>` or `/../<badi>` (choose any; comply with namespace)
Interface	`IF_EX_<badi>` or `ZIF_EX_<badi>` or `/../IF_EX_<badi>` (choose any; comply with namespace)
Methods	Choose any name you want
Generated Business Add-In Class (Adapter Class)	`CL_EX_<badi>` or `ZCL_EX_<badi>` or `/../CL_EX_<badi>` (cannot be changed)

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Finding a Business Add-In

- **Directly in application:**

System -> Status
-> Double-click on
program name
Find character string…

CL_EXITHANDLER

- `Global in program <prog>`

→ Double-click reference variable

Double-click Interface Class Builder:

→ Where-used list for interface

→ CL_EX_<Badl-Name>

→ SE18:<BA-Name>

→ Documentation

- **Using tools**

Application hierarchy

Repository Information System

IMG

→ Documentation

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BADI Implementation: Naming Conventions

BADI Implementation	`<impl>` or `Z<impl>` or `/../<impl>` (choose any; comply with namespace)
Interface	`IF_EX_<badi>` or `ZIF_EX_<badi>` or `/../IF_EX_<badi>` (defined in business add-in definition)
Methods	Defined in business add-in definition
Implementing Class	`CL_IM_<impl>` or `ZCL_IM_<impl>` or `/../CL_IM_<impl>` (choose any; comply with namespace)

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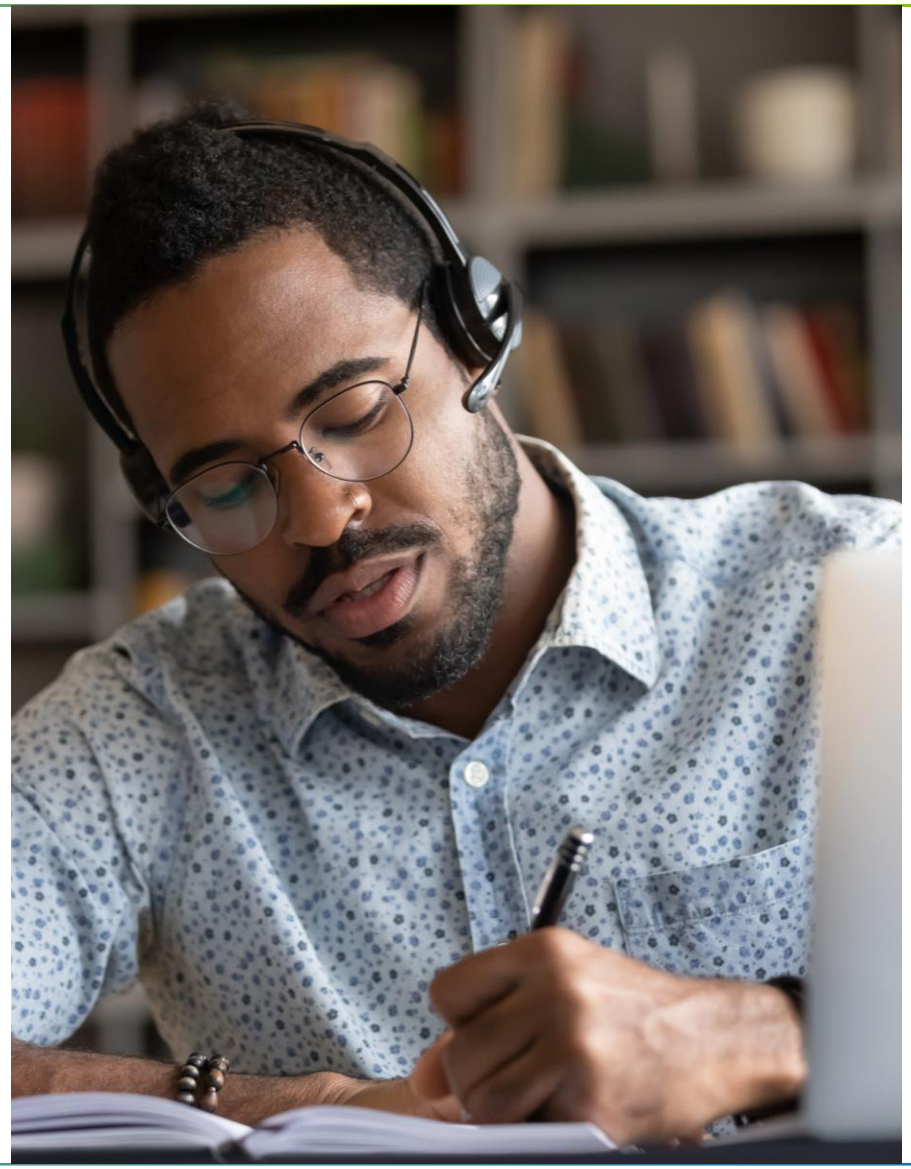
Hands-On Labs



Instructions

- Specify the Customer Number as — <Trainee ID>-<00 to 99>

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Hands-On Activity 5

Activity Details: BADI Implementation with Delivery (VL02N)	
Problem Statement	Find the BADI definition you would use, and ensure that on save of OUTBOUND delivery, the following are saved in a custom table. Implement the BADI. • reference document number, • ref document item number, • ref document material number, • quantity and • UOM Note that the code should be executed only for reference document type (VGTYP) = C.

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Hands-On Activity 6

Activity Details: BADI Implementation with the shipment (VT02N)	
Problem Statement	Find the BADI definition that you can be used to check the below fields while saving the shipping document (VT02N). Implement the BADI. • Field Container ID cannot be left blank • Field External ID 1 cannot be left blank • If External ID 1 is not blank, copy the value in External ID 2, which should be saved in the VTTK table Give an error message on validation failure. Ensure that the screen fields are not disabled on issuing the error message.

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Summary

Here are the key learning points of the module.

- Modifications are changes to SAP repository objects made at the customer site. If SAP delivers a changed version of the object, the customer's system must be adjusted to reflect these changes.
- Business transaction events are enhancements that were developed for the financial accounting component of R/3.
- In product function, the product groups enhancements together and it can be activated or deactivated as a whole.
- The SAP application programmer creates SAP enhancements from function module exits, menu exits, and screen exits.
- Screen exits are a kind of customer exit. They allow displaying additional objects in an SAP application program screen.
- Menu enhancements permit you to add additional menu entries to an SAP standard menu.
- Application hierarchy, repository information system, and IMG are the tools used in finding a business add-in.

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